

Exact connectedness thresholds and strict order monotonicity for complex tridiagonal Toeplitz pseudospectra

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Abstract

We establish three connectedness results for open Euclidean-norm pseudospectra. First, for a finite matrix whose spectrum lies on an affine line, a natural least-singular-value contraction condition forces every pseudospectral component to be contractible and reduces connectedness to a one-dimensional barrier. Second, this principle yields an exact threshold for the full complex tridiagonal Toeplitz family $T_n(\alpha, d, \beta) = \text{tridiag}(\alpha, d, \beta)$. Set $c = \max\{|\alpha|, |\beta|\}$. When $c > 0$, unitary similarity and affine transformations reduce the problem to $A_n(a) = \text{tridiag}(a, 0, 1)$, with $a = \min\{|\alpha|, |\beta|\}/c$. In the irreducible nonnormal regime $0 < a < 1$, if $\gamma_n(a)$ is the real-axis least-singular-value barrier, then $\sigma_\varepsilon(T_n)$ is connected exactly when $\varepsilon > c\gamma_n(a)$. Third, the thresholds obey the strict all-order inequality $\gamma_{n+1}(a) < \gamma_n(a)$ for every $n \geq 2$. Thus connectedness persists with increasing order. Matching bounds of order $(n+1)a^{n/2}$ give a two-term logarithmic asymptotic for the first connected order. The normal boundary has an algebraic threshold, while the one-sided and scalar cases have zero threshold.

Keywords: pseudospectrum, tridiagonal Toeplitz matrix, nonnormal matrix, least singular value, pseudospectral connectedness, matrix-order monotonicity.

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1 Introduction

Let $n \geq 1$, $A \in \mathbb{C}^{n \times n}$, and $\varepsilon > 0$. The open Euclidean-norm ε -pseudospectrum is

$$\sigma_\varepsilon(A) = \sigma(A) \cup \{z \in \mathbb{C} \setminus \sigma(A) : \|(zI_n - A)^{-1}\|_2 > \varepsilon^{-1}\} = \{z \in \mathbb{C} : s_{\min}(zI_n - A) < \varepsilon\}. \quad (1.1)$$

Here and throughout, I_n is the n -by- n identity, $s_{\min}$ denotes the least singular value, and $\|\cdot\|_2$ is the Euclidean norm on vectors or its induced operator norm on matrices. Equivalently, (1.1) is the union of the spectra of $A + \Delta A$ over all complex perturbations satisfying $\|\Delta A\|_2 < \varepsilon$ [22]. Pseudospectral components therefore encode spectral clusters stable under unstructured perturbations. Their merger is also related to distance to defectivity: for matrices with distinct eigenvalues, first-coalescence points are the lowest generalized saddle points of the least-singular-value function, and their common level equals the distance to the set of defective matrices in the induced Euclidean operator norm [1]. Thus connectedness depends on the global geometry of the least-singular-value landscape, not on the spectrum alone.

The contribution of this paper is threefold.

1. *A general spectral-line contraction theorem.* For a finite matrix with spectrum on an affine line, a natural contraction condition on least-singular-value sublevel sets makes every pseudospectral component contractible and reduces connectedness to the largest one-dimensional barrier between adjacent eigenvalues.

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2. *An exact Toeplitz threshold.* For every $n \geq 2$ and all complex α, d, β , we give a necessary-and-sufficient connectedness criterion for $T_n(\alpha, d, \beta) = \text{tridiag}(\alpha, d, \beta)$, including its reducible, irreducible nonnormal, and normal regimes.
3. *Strict monotonicity at every order.* In the canonical nonnormal regime $0 < a < 1$, the exact thresholds satisfy $\gamma_{n+1}(a) < \gamma_n(a)$ for all $n \geq 2$. Hence connectedness, once attained, persists at every larger order.

We study the full complex tridiagonal Toeplitz family

$$T_n(\alpha, d, \beta) = \text{tridiag}(\alpha, d, \beta) = \begin{bmatrix} d & \beta & & \\ \alpha & d & \ddots & \\ & \ddots & \ddots & \beta \\ & & \alpha & d \end{bmatrix}, \quad n \geq 2, \quad \alpha, d, \beta \in \mathbb{C}, \quad (1.2)$$

where the arguments of tridiag are the subdiagonal, diagonal, and superdiagonal. Its Euclidean-norm pseudospectral geometry is determined by the two real quantities

$$c = \max\{|\alpha|, |\beta|\}, \quad a = \frac{\min\{|\alpha|, |\beta|\}}{c} \in [0, 1],$$

when $c > 0$. A diagonal unitary, possibly followed by reversal, reduces $T_n(\alpha, d, \beta)$ to a translate, rotation, and positive multiple of the canonical matrix

$$A_n(a) = \text{tridiag}(a, 0, 1). \quad (1.3)$$

The parameter in (1.3) separates three regimes: $a = 0$ is the one-sided nilpotent-shift boundary, $0 < a < 1$ is the irreducible nonnormal regime, and $a = 1$ is the normal boundary. The scalar case $c = 0$ is treated separately.

For $0 < a < 1$, write

$$\gamma_n(a) = \max_{|x| \leq 2\sqrt{a} \cos(\pi/(n+1))} s_{\min}(xI_n - A_n(a)). \quad (1.4)$$

Then $\sigma_\varepsilon(T_n(\alpha, d, \beta))$ is connected if and only if $\varepsilon > c\gamma_n(a)$.

The proof compares global barriers on spectral intervals that vary with the matrix order. Diagonal similarity identifies the spectrum but has Euclidean condition number $a^{-(n-1)/2}$; interlacing is pointwise, while the barrier maximizes over moving gaps whose central geometry changes with parity. Toeplitz asymptotics describe the eventual scale, whereas adjacent-order monotonicity requires a finite-order comparison.

The Toeplitz background begins with the Schmidt–Spitzer characterization of the eigenvalue limit set for matrices generated by a Laurent polynomial [19]. Reichel and Trefethen analyze finite non-Hermitian Toeplitz eigenvalues, resolvent norms, and pseudospectra [18]. For large truncated Wiener–Hopf operators, Böttcher establishes limiting results for inverse norms, pseudospectra, and singular values [6]; a book-length treatment of banded Toeplitz spectral theory, including resolvents, pseudospectra, singular values, and structured perturbations, is given in [8]. Closed-form spectral, sensitivity, and pseudospectral properties of tridiagonal Toeplitz matrices are developed in [17], while algorithms for Toeplitz-structured pseudospectra and their extreme points are studied in [11].

Recent directions include convergent spectral and pseudospectral inclusion sets for banded and band-dominated operators [12] and individual asymptotic expansions for all eigenvalues in a non-Hermitian tetradiagonal Toeplitz class [5]. Spectra and pseudospectra of tridiagonal k -Toeplitz operators and matrices are studied in [3]; the corrected finite-matrix pseudospectral limit is given in [4].

At the level of topology, pseudospectral optimization and component geometry are developed in [9]. Pseudospectral growth is related to spectral conditioning, and the first loss of n components is shown to equal the distance to matrices with multiple eigenvalues in [10]. Alam, Bora, Byers, and Overton further identify first-coalescence points as lowest generalized saddle points and relate their level to distance to defectivity [1].

A parallel line of work quantifies eigenvalue instability in random non-self-adjoint matrices [13], analyzes pseudospectra, exponential resolvent growth, and localization in random bidiagonal models [21], and exhibits recent Toeplitz-like examples of spectral instability [7]. In non-Hermitian lattice problems, Ammari et al. prove exponential edge localization for the nontrivial eigenvectors of a finite gauge-capacitance chain [2]; Kiorpelidis and Makris identify a critical order beyond which the initially separate pseudospectral clouds form a single cloud [16]; and Sirker uses pseudospectra to separate nonnormal spectral instability and the skin effect from stable topological information [20].

Together these works provide limiting sets, computable enclosures, structured perturbation models, eigenvalue asymptotics, and general coalescence principles. The present result complements them with an exact finite-order connectedness criterion for the full complex tridiagonal Toeplitz family and a strict adjacent-order comparison of its Euclidean-norm barriers, organized by the spectral-line contraction theorem.

For $0 < a < 1$, the threshold satisfies

$$\gamma_n(a) \asymp_a (n+1)a^{n/2}, \quad (1.5)$$

with an optimal factor $n+1$; these bounds yield a two-term logarithmic asymptotic for the first connected order. At $a = 1$, the barrier is half the largest adjacent eigenvalue gap and has order $(n+1)^{-1}$; at $a = 0$, it vanishes. Scalar matrices also have zero threshold. Thus the nonnormal interior, normal boundary, and zero-threshold cases have exponential–polynomial, algebraic, and zero scales, respectively.

Section 2 proves this topological principle, and Section 3 performs the unitary–affine reduction and states the classification. The Toeplitz-specific proof occupies Sections 4 to 6: vertical monotonicity gives the real-axis criterion, signed-pencil and chord arguments prove strict matrix-order comparison, and rectangular factorizations give the two-sided bounds. The algebraic certificates are collected in appendices A to F; numerical illustrations and broader implications appear in Sections 7 and 8.

2 A vertical-monotonicity principle

This section isolates the topology from the Toeplitz calculation. Let $n \geq 1$, let $B \in \mathbb{C}^{n \times n}$ have real spectrum, and put

$$h_B(x, y) = s_{\min}((x + iy)I_n - B), \quad x, y \in \mathbb{R}.$$

We say that B has the *vertical contraction property* if

$$h_B(x, ty) \leq h_B(x, y) \quad (x, y \in \mathbb{R}, 0 \leq t \leq 1). \quad (2.1)$$

For a real matrix, (2.1) holds whenever $y \mapsto h_B(x, y)$ is nondecreasing on $[0, \infty)$, since singular values are invariant under complex conjugation.

Write

$$\lambda_-(B) = \min \sigma(B), \quad \lambda_+(B) = \max \sigma(B),$$

and define the real-axis barrier

$$\Gamma(B) = \max_{\lambda_-(B) \leq x \leq \lambda_+(B)} s_{\min}(xI_n - B). \quad (2.2)$$

The maximum exists by continuity and equals the largest pointwise backward error: for each x ,

$$s_{\min}(xI_n - B) = \min\{\|\Delta B\|_2 : x \in \sigma(B + \Delta B)\}. \quad (2.3)$$

Identity (2.3) follows because, if $(xI_n - B - \Delta B)v = 0$ for a unit vector v , then $s_{\min}(xI_n - B) \leq \|\Delta B\|_2$. Conversely, unit singular vectors satisfying $(xI_n - B)v = s_{\min}(xI_n - B)u$ yield the minimizing rank-one perturbation $\Delta B = s_{\min}(xI_n - B)uv^*$.

We use the following standard component fact in a form that applies to every finite matrix [9, 22].

Lemma 2.1 (Eigenvalue in every component). *Let $n \geq 1$, $M \in \mathbb{C}^{n \times n}$, and $\varepsilon > 0$. Every connected component of $\sigma_\varepsilon(M)$ contains an eigenvalue of M .*

Proof. The estimate $s_{\min}(zI_n - M) \geq |z| - \|M\|_2$ makes the pseudospectrum bounded. Let C be a component containing no eigenvalue. If an eigenvalue belonged to $\overline{C}$, continuity of the least singular value would give a pseudospectral ball meeting C , and maximality would put that eigenvalue in C . Thus $\overline{C}$ lies in the resolvent set. On a neighborhood of this compact set, $z \mapsto \|(zI_n - M)^{-1}\|_2$ is continuous and subharmonic. It is at most ε^{-1} on ∂C , whereas it is greater than ε^{-1} in C . The maximum principle gives a contradiction. $\square$

Theorem 2.2 (Vertical topology theorem). *Let $n \geq 1$. If $B \in \mathbb{C}^{n \times n}$ has real spectrum and satisfies (2.1), then, for every $\varepsilon > 0$, each connected component of $\sigma_\varepsilon(B)$ is contractible, and*

$$\sigma_\varepsilon(B) \text{ is connected} \iff [\lambda_-(B), \lambda_+(B)] \subset \sigma_\varepsilon(B) \iff \varepsilon > \Gamma(B). \quad (2.4)$$

If $\Gamma(B) > 0$, then at the positive level $\varepsilon = \Gamma(B)$, a maximizing real point is excluded from the open pseudospectrum.

Proof. Put $\Omega = \sigma_\varepsilon(B)$. By (2.1),

$$\mathcal{H}_t(x + iy) = x + i(1 - t)y, \quad 0 \leq t \leq 1,$$

maps Ω into itself. For any component C , the path $t \mapsto \mathcal{H}_t(z)$ cannot leave C . Since $\mathcal{H}_t$ fixes the real axis, $\mathcal{H}_1(C) = C \cap \mathbb{R}$, and $\mathcal{H}_t|_C$ is a strong deformation retraction onto this intersection. It is nonempty by lemma 2.1; it is connected as the continuous image $\mathcal{H}_1(C)$, and is therefore an interval. Thus C is contractible.

If Ω is connected, then $\Omega \cap \mathbb{R}$ is an interval containing all eigenvalues and hence their convex hull. Conversely, if that convex hull lies in Ω , it belongs to one component and contains every eigenvalue. By lemma 2.1, no other component can exist. This proves the first equivalence in (2.4); the second is exactly the strict sublevel definition and (2.2). $\square$

Translation, rotation, and positive scaling translate, rotate, and scale the pseudospectrum and its barrier. The theorem therefore extends from the real axis to any affine spectral line and applies to every matrix in (1.2).

3 Complex tridiagonal Toeplitz matrices and main results

We first normalize the phases, orientation, diagonal, and overall scale in (1.2). Let $V_n = \text{tridiag}(0, 0, 1)$, let J_n be the reversal permutation matrix, and, for $|q| = 1$, put

$$D_n(q) = \text{diag}(1, q, \dots, q^{n-1}).$$

Proposition 3.1 (Unitary–affine reduction). *Let $n \geq 2$, $c = \max\{|\alpha|, |\beta|\} > 0$, and $a = \min\{|\alpha|, |\beta|\}/c$. There are a unitary matrix S_n and a unimodular number ω such that*

$$S_n^* T_n(\alpha, d, \beta) S_n = dI_n + c\omega A_n(a). \quad (3.1)$$

Consequently, for every $\varepsilon > 0$,

$$\sigma_\varepsilon(T_n(\alpha, d, \beta)) = d + c\omega \sigma_{\varepsilon/c}(A_n(a)). \quad (3.2)$$

Proof. First suppose $0 < |\alpha| \leq |\beta|$. Write $u_\alpha = \alpha/|\alpha|$ and $u_\beta = \beta/|\beta|$. Choose $|q| = 1$ with $q^2 = u_\alpha \overline{u_\beta}$, and put $\omega = u_\beta q = u_\alpha q^{-1}$. Direct multiplication gives

$$D_n(q)^* T_n(\alpha, d, \beta) D_n(q) = dI_n + |\beta|\omega A_n(|\alpha|/|\beta|).$$

If $|\alpha| > |\beta| > 0$, first conjugate by J_n , which interchanges the two off-diagonals, and apply the preceding case. If $\alpha = 0$ and $\beta \neq 0$, take $S_n = I_n$ and $\omega = \beta/|\beta|$; if $\beta = 0$ and $\alpha \neq 0$, take $S_n = J_n$ and $\omega = \alpha/|\alpha|$. Applying the singular-value transformation in (3.1) gives (3.2), because unitary conjugation preserves singular values, while

$$\sigma_\varepsilon(dI_n + c\omega A) = d + c\omega \sigma_{\varepsilon/c}(A).$$

□

The boundary thresholds follow directly. Define

$$\eta_n = \begin{cases} 2 \sin \frac{\pi}{2(n+1)}, & n \text{ even}, \\ \sin \frac{\pi}{n+1}, & n \text{ odd}. \end{cases} \quad (3.3)$$

Proposition 3.2 (Reducible and normal boundaries). *Let $n \geq 2$ and $\varepsilon > 0$.*

1. *If $\alpha = \beta = 0$, then $\sigma_\varepsilon(T_n) = \{z : |z - d| < \varepsilon\}$. If exactly one off-diagonal is nonzero, then $\sigma_\varepsilon(T_n)$ is also an open disk centered at d . In both cases it is connected and contractible for every n ; its connectedness threshold is zero.*
2. *If $|\alpha| = |\beta| = c > 0$, then every pseudospectral component is contractible and*

$$\sigma_\varepsilon(T_n) \text{ is connected} \iff \varepsilon > c\eta_n. \quad (3.4)$$

Moreover, $\eta_{n+1} < \eta_n$ for every $n \geq 2$, and

$$0 \leq \frac{\pi}{n+1} - \eta_n \leq \frac{\pi^3}{6(n+1)^3}. \quad (3.5)$$

In particular,

$$\eta_n = \frac{\pi}{n+1} + O(n^{-3}). \quad (3.6)$$

Proof. The scalar case follows directly. If exactly one off-diagonal is nonzero, proposition 3.1 reduces the matrix to the nilpotent Jordan shift V_n , followed by an affine transformation of the spectral plane. By lemma 2.1, every component of $\sigma_\varepsilon(V_n)$ contains its only eigenvalue, zero; hence the pseudospectrum is connected. Moreover, $D_n(q)^* V_n D_n(q) = qV_n$, while unitary invariance and $\sigma_\varepsilon(qV_n) = q\sigma_\varepsilon(V_n)$ show that it is invariant under every rotation. Its set of attained moduli is therefore a connected interval containing zero. Boundedness and openness make this interval $[0, R_{n,\varepsilon})$ for some $R_{n,\varepsilon} > 0$, and thus $\sigma_\varepsilon(V_n) = \{z : |z| < R_{n,\varepsilon}\}$. The affine transformation in (3.2) gives the asserted disk centered at d .

In part 2, proposition 3.1 reduces the problem to the real symmetric matrix $A_n(1)$, whose eigenvalues are $2 \cos(k\pi/(n+1))$, $1 \leq k \leq n$. Its open pseudospectrum is the union of open disks of radius ε/c about

these collinear points. It is connected exactly when twice that radius exceeds every adjacent gap. The identity

$$\cos \frac{k\pi}{n+1} - \cos \frac{(k+1)\pi}{n+1} = 2 \sin \frac{(2k+1)\pi}{2(n+1)} \sin \frac{\pi}{2(n+1)}$$

shows that the varying sine factor has maximum 1 for even n and $\cos(\pi/(2(n+1)))$ for odd n . Hence half the largest gap is η_n in (3.3). At equality the two relevant open disks do not meet, so the inequality in (3.4) is strict. For this normal matrix,

$$s_{\min}((x + iy)I_n - A_n(1)) = \min_k \sqrt{(x - \lambda_k)^2 + y^2},$$

so the vertical contraction property follows. Component contractibility follows from Theorem 2.2 and is preserved by the affine transformation in (3.2).

The two parity transitions are strict. From $2m$ to $2m+1$, if $x = \pi/(4m+2)$ and $y = \pi/(2m+2)$, then $y < 2x < \pi/2$, so $\sin y < \sin(2x) < 2 \sin x$. From $2m+1$ to $2m+2$, put $p = m+1 \geq 2$, $x = \pi/(2p)$, and $y = \pi/(4p+2)$. The estimates $\sin x > x - x^3/6$, $\sin y < y$, and $24p^2 > \pi^2(2p+1)$ give $\sin x > 2 \sin y$; the last inequality follows because $24p^2/(2p+1)$ is increasing for $p \geq 2$ and equals $96/5 > \pi^2$ at $p = 2$. Finally, $0 \leq x - \sin x \leq x^3/6$ for $x \geq 0$. Applied to the two parity formulas in (3.3), this gives (3.5), and hence (3.6). $\square$

For the nonnormal regime, fix $0 < a < 1$, put $r = \sqrt{a}$, and write $A_n = A_n(a) = V_n + aV_n^T$. Define

$$\mathcal{G}_n(\varrho) = \text{diag}(1, \varrho, \dots, \varrho^{n-1}).$$

The diagonal similarity

$$A_n(a) = \mathcal{G}_n(r) r(V_n + V_n^T) \mathcal{G}_n(r)^{-1} \quad (3.7)$$

gives the displayed real spectrum, but is nonunitary for $0 < a < 1$, with Euclidean condition number $\|\mathcal{G}_n(r)\|_2 \|\mathcal{G}_n(r)^{-1}\|_2 = r^{-(n-1)}$. Define

$$\begin{aligned} I_n^{\text{sp}} &= \left[-2r \cos \frac{\pi}{n+1}, 2r \cos \frac{\pi}{n+1} \right], & g_n(x) &= s_{\min}(xI_n - A_n), \\ \gamma_n(a) &= \max_{x \in I_n^{\text{sp}}} g_n(x), & N_c(a, \varepsilon) &= \min\{n \geq 2 : \sigma_\varepsilon(A_n(a)) \text{ is connected}\}. \end{aligned}$$

The theorem below shows that the minimum defining N_c exists. When a is fixed, its argument is suppressed. Define the explicit lower and upper barriers

$$\mathcal{U}_n(a) = r^n \csc \frac{\pi}{n+1}, \quad (3.8)$$

$$\mathcal{L}_n(a) = \left(1 + r^2 - 2r \cos \frac{\pi}{n+1} \right) \frac{r^n(1-r^2)}{(1-r^{2(n+1)}) \sin \frac{3\pi}{2(n+1)}}. \quad (3.9)$$

Theorem 3.3 (Canonical nonnormal theorem). *For every $0 < a < 1$ and $\varepsilon > 0$, the following assertions hold.*

1. *For every $n \geq 2$, every component of $\sigma_\varepsilon(A_n(a))$ is contractible.*
2. *The barrier sequence satisfies $\gamma_{n+1}(a) < \gamma_n(a)$ for every $n \geq 2$, and $\gamma_n(a) \rightarrow 0$ as $n \rightarrow \infty$. Moreover, for every $n \geq 2$,*

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff \varepsilon > \gamma_n(a).$$

3. *Connectedness persists after its first occurrence:*

$$\{n \geq 2 : \sigma_\varepsilon(A_n(a)) \text{ is connected}\} = \{n \geq N_c(a, \varepsilon)\}.$$

Moreover,

$$1 + \max\left(\{1\} \cup \{n \geq 2 : \mathcal{L}_n(a) \geq \varepsilon\}\right) \leq N_c(a, \varepsilon), \quad (3.10)$$

$$N_c(a, \varepsilon) \leq \min\{n \geq 2 : \mathcal{U}_n(a) < \varepsilon\}.$$

4. For fixed a , as $\varepsilon \downarrow 0$,

$$N_c(a, \varepsilon) = \frac{2}{\log(1/a)} (\log(1/\varepsilon) + \log\log(1/\varepsilon)) + O_a(1), \quad (3.11)$$

and $N_c(a, \varepsilon) = 2$ if and only if $\varepsilon > a$.

Define the full-family threshold

$$\Theta_n(\alpha, \beta) = \begin{cases} 0, & \alpha\beta = 0, \\ c\gamma_n(a), & \alpha\beta \neq 0 \text{ and } |\alpha| \neq |\beta|, \\ c\eta_n, & |\alpha| = |\beta| = c > 0, \end{cases} \quad (3.12)$$

where the middle case uses $c = \max\{|\alpha|, |\beta|\}$ and $a = \min\{|\alpha|, |\beta|\}/c$. Also set

$$N_T(\alpha, d, \beta; \varepsilon) = \min\{n \geq 2 : \sigma_\varepsilon(T_n(\alpha, d, \beta)) \text{ is connected}\}.$$

The classification below shows that this minimum exists. By proposition 3.1, it is independent of d and of the phases of the off-diagonal entries. Whenever $\alpha\beta \neq 0$, write

$$c = \max\{|\alpha|, |\beta|\}, \quad a = \frac{\min\{|\alpha|, |\beta|\}}{c} \in (0, 1].$$

Theorem 3.4 (Complex tridiagonal Toeplitz classification). *Let $\alpha, d, \beta \in \mathbb{C}$ and $\varepsilon > 0$.*

1. *For every $n \geq 2$, every component of $\sigma_\varepsilon(T_n(\alpha, d, \beta))$ is contractible, and*

$$\sigma_\varepsilon(T_n(\alpha, d, \beta)) \text{ is connected} \iff \varepsilon > \Theta_n(\alpha, \beta). \quad (3.13)$$

2. *If $\alpha\beta \neq 0$, then $\Theta_{n+1}(\alpha, \beta) < \Theta_n(\alpha, \beta)$ for every $n \geq 2$, and $\Theta_n(\alpha, \beta) \rightarrow 0$ as $n \rightarrow \infty$. Hence connectedness persists after its first occurrence. If $\alpha\beta = 0$, the threshold is identically zero and the pseudospectrum is connected at every order.*

3. *For fixed α, d, β , in the irreducible nonnormal case $0 < a < 1$,*

$$N_T(\alpha, d, \beta; \varepsilon) = N_c(a, \varepsilon/c) = \frac{2}{\log(1/a)} \left(\log \frac{c}{\varepsilon} + \log\log \frac{c}{\varepsilon} \right) + O_a(1) \quad (3.14)$$

as $\varepsilon \downarrow 0$. In the normal case $a = 1$,

$$N_T(\alpha, d, \beta; \varepsilon) = \frac{\pi c}{\varepsilon} + O(1). \quad (3.15)$$

If $\alpha\beta = 0$, then $N_T(\alpha, d, \beta; \varepsilon) = 2$.

Proof. For $c > 0$, (3.13) follows from proposition 3.1 and the definition (3.12). The nonnormal assertions follow from Theorem 3.3, with ε replaced by ε/c ; the boundary assertions follow from proposition 3.2. In the normal case, put $m = N_T(\alpha, d, \beta; \varepsilon)$. For sufficiently small ε , one has $m > 2$ and

$$c\eta_{m-1} \geq \varepsilon > c\eta_m.$$

The bound (3.5) first gives $m \sim \pi c/\varepsilon$, and then $\varepsilon = \pi c/m + O(m^{-2})$. Hence $m = \pi c/\varepsilon + O(1)$, proving (3.15). The nonnormal asymptotic in (3.14) is the corresponding scaled form of (3.11). If $c = 0$, the scalar case applies directly. $\square$

4 Verification of vertical monotonicity for $A_n(a)$

For the canonical family, the topological reduction rests on a monotonicity property that makes every merger of pseudospectral components visible on the real axis.

Lemma 4.1 (Vertical monotonicity). *Let $n \geq 2$ and $0 < a < 1$. For every real x ,*

$$y \mapsto s_{\min}((x + iy)I_n - A_n), \quad y \geq 0,$$

is nondecreasing.

Proof. Put $Y = y^2$, and, for $t \geq 0$, define

$$\begin{aligned} \mathcal{P}_n(Y, t) &= ((x + i\sqrt{Y})I_n - A_n)^*((x + i\sqrt{Y})I_n - A_n) - t^2 I_n, \\ F_n(Y, t) &= \det \mathcal{P}_n(Y, t). \end{aligned}$$

Proposition A.1 proves the finite-dimensional implication

$$\mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0. \quad (4.1)$$

Finally fix $Y_0 \geq 0$. When the least singular value at Y_0 equals zero, monotonicity follows from nonnegativity of singular values. Otherwise take $0 \leq t < s_{\min}((x + i\sqrt{Y_0})I_n - A_n)$. On the maximal interval to the right of Y_0 on which $\mathcal{P}_n(Y, t) > 0$, implication (4.1) makes its determinant strictly increasing. At any finite right endpoint, continuity of Hermitian eigenvalues would produce a positive-semidefinite singular matrix with determinant zero. Strict increase from the positive determinant at Y_0 precludes such an endpoint. Thus $\mathcal{P}_n(Y, t) > 0$ for every $Y \geq Y_0$. Letting t increase to the least singular value at Y_0 proves the lemma. $\square$

Because $A_n(a)$ is real, lemma 4.1 gives the vertical contraction property (2.1). Its spectrum is the real set $\{2\sqrt{a} \cos(k\pi/(n+1)) : 1 \leq k \leq n\}$, whose convex hull is I_n^{sp} . Applying Theorem 2.2 therefore gives contractibility of every component and

$$\sigma_\varepsilon(A_n(a)) \text{ is connected} \iff I_n^{\text{sp}} \subset \sigma_\varepsilon(A_n(a)) \iff \gamma_n(a) < \varepsilon. \quad (4.2)$$

At $\varepsilon = \gamma_n(a)$, a maximizing real point is excluded from the open pseudospectrum.

5 Strict matrix-order comparison of real-gap barriers

The global barrier may occur in any inter-eigenvalue gap, while the central gap changes with parity. We therefore compare every gap. Persymmetry selects the signed eigenvalue branch whose magnitude is the least singular value; folded Chebyshev coordinates and chord geometry compare the noncentral gaps, and a separate argument treats the parity-changing central gaps. Write the eigenvalues in decreasing order as

$$\lambda_{n,k} = 2r \cos \frac{k\pi}{n+1}, \quad 1 \leq k \leq n,$$

and call

$$G_{n,k} = [\lambda_{n,k+1}, \lambda_{n,k}], \quad \gamma_{n,k} = \max_{x \in G_{n,k}} g_n(x), \quad 1 \leq k < n,$$

the closed spectral gaps and their heights. Continuity guarantees that each maximum is attained. The spectral endpoints have height zero, whereas $g_n(x) > 0$ in every open gap. Thus every gap height is positive, every maximizer lies in the open gap, and

$$\gamma_n = \max_{1 \leq k < n} \gamma_{n,k}. \quad (5.1)$$

Lemma 5.1 (Strict barrier comparison). *For every $n \geq 2$,*

$$\gamma_{n+1} < \gamma_n.$$

5.1 Signed-pencil representation and branch selection

Recall from Section 3 that J_n is the reversal matrix. Persymmetry supplies the signed spectral representation needed for the matrix-order comparison: $A_n J_n = J_n A_n^T$, so $B_n(x) = (xI_n - A_n)J_n$ is real symmetric. Moreover,

$$(xI_n - A_n - sJ_n)J_n = B_n(x) - sI_n.$$

Since J_n is orthogonal, the absolute values of the eigenvalues of $B_n(x)$ are exactly the singular values of $xI_n - A_n$, including multiplicity. Consequently,

$$s_{\min}(xI_n - A_n) = \min\{|s| : \det(xI_n - A_n - sJ_n) = 0\}. \quad (5.2)$$

Set $q_n(x, s) = \det(xI_n - A_n - sJ_n)$; the minimum in (5.2) selects its signed branch nearest zero.

Lemma 5.2 (The middle signed branch). *Write the eigenvalues of $B_n(x)$ in decreasing order as $\mu_1(x) \geq \dots \geq \mu_n(x)$, and abbreviate $\lambda_k = \lambda_{n,k}$. If $\lambda_{j+1} < x < \lambda_j$, then*

$$s_{\min}(xI_n - A_n) = |s_*(x)|, \quad \text{sgn}(s_*(x)) = (-1)^j,$$

where

$$s_*(x) = \begin{cases} \mu_m(x), & n = 2m, \\ \mu_{m+1}(x), & n = 2m + 1. \end{cases}$$

Proof. Let $A_n^{\text{sym}} = r(V_n + V_n^T)$. The identities

$$\begin{aligned} A_n &= \mathcal{G}_n(r) A_n^{\text{sym}} \mathcal{G}_n(r)^{-1}, \\ \mathcal{G}_n(r)^{-1} J_n &= r^{-(n-1)} J_n \mathcal{G}_n(r) \end{aligned}$$

give

$$B_n(x) = r^{-(n-1)} \mathcal{G}_n(r) (xI_n - A_n^{\text{sym}}) J_n \mathcal{G}_n(r). \quad (5.3)$$

The sine eigenvector of A_n^{sym} belonging to λ_k is an eigenvector of J_n with eigenvalue $(-1)^{k+1}$. Hence (5.3) and Sylvester's law of inertia show that $B_n(x)$ has the same inertia as the diagonal matrix with entries

$$(-1)^{k+1}(x - \lambda_k), \quad 1 \leq k \leq n.$$

For a real symmetric matrix H , let $\nu_+(H)$ denote the number of its positive eigenvalues, counted with multiplicity. Counting signs gives

$$\nu_+(B_{2m}(x)) = \begin{cases} m, & j \text{ even}, \\ m-1, & j \text{ odd}, \end{cases} \quad \nu_+(B_{2m+1}(x)) = \begin{cases} m+1, & j \text{ even}, \\ m, & j \text{ odd}. \end{cases} \quad (5.4)$$

First let $n = 2m$. Permute the coordinates into the two groups

$$(1, \dots, m), \quad (2m, 2m-1, \dots, m+1).$$

Here $\mathbf{e}_j$ denotes the j th Euclidean coordinate vector in the ambient space. Put $P_m(x) = xI_m - A_m$. In these coordinates

$$\begin{aligned} B_{2m}(x) &= \widehat{P}_m(x) - \mathcal{R}_{\text{bd}}, \\ \widehat{P}_m(x) &= \begin{bmatrix} 0 & P_m(x) \\ P_m(x)^T & 0 \end{bmatrix}, \\ \mathcal{R}_{\text{bd}} &= \begin{bmatrix} \mathbf{e}_m \mathbf{e}_m^T & 0 \\ 0 & a \mathbf{e}_m \mathbf{e}_m^T \end{bmatrix}, \end{aligned} \quad (5.5)$$

In (5.5), $\mathcal{R}_{\text{bd}} \geq 0$ and $\text{rank } \mathcal{R}_{\text{bd}} = 2$. For $\mathbf{S} = \text{diag}(I_m, -I_m)$,

$$\widetilde{B} = -\mathbf{S}B_{2m}(x)\mathbf{S} = \widehat{P}_m(x) + \mathcal{R}_{\text{bd}} = B_{2m}(x) + 2\mathcal{R}_{\text{bd}}.$$

If $\widetilde{\mu}_1 \geq \cdots \geq \widetilde{\mu}_{2m}$ are the eigenvalues of $\widetilde{B}$, then

$$\widetilde{\mu}_k = -\mu_{2m+1-k}. \quad (5.6)$$

Combining the reflection (5.6) with positive-semidefinite rank-two interlacing gives

$$\widetilde{\mu}_k \geq \mu_k \geq \widetilde{\mu}_{k+2}, \quad 1 \leq k \leq 2m-2. \quad (5.7)$$

If j is even, the only candidates for minimum modulus have moduli μ_m and $-\mu_{m+1} = \widetilde{\mu}_m$, and $\mu_m \leq \widetilde{\mu}_m$. If j is odd and $m \geq 2$, the two candidates have moduli μ_{m-1} and $-\mu_m = \widetilde{\mu}_{m+1}$; taking $k = m-1$ in (5.7) gives $\mu_{m-1} \geq \widetilde{\mu}_{m+1}$. For $m = 1$, both eigenvalues in the sole gap are negative, so the larger one μ_1 has minimum modulus. Thus μ_m always attains the minimum modulus, and its sign follows from (5.4). Equality in an interlacing comparison corresponds to a multiple least singular value.

Now let $n = 2m + 1$. Group the first and last m coordinates as above and leave the middle coordinate last. Then

$$B_{2m+1}(x) = \begin{bmatrix} \widehat{P}_m(x) & \mathbf{w} \\ \mathbf{w}^T & x \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -\mathbf{e}_m \\ -a\mathbf{e}_m \end{bmatrix}. \quad (5.8)$$

In an open spectral gap of A_{2m+1} , $P_m(x)$, and hence the principal block $\widehat{P}_m(x)$ in (5.8), is nonsingular. Indeed, the explicit path spectrum gives

$$\sigma(A_m) = \left\{ 2r \cos \frac{\ell\pi}{m+1} : 1 \leq \ell \leq m \right\} = \left\{ \lambda_{2m+1, 2\ell} : 1 \leq \ell \leq m \right\},$$

so its zeros are exactly the even-indexed eigenvalues of A_{2m+1} , none of which lies in an open spectral gap. If $s_1(P_m) \geq \cdots \geq s_m(P_m)$ are the singular values of $P_m(x)$, then the eigenvalues of the principal submatrix $\widehat{P}_m(x)$ are

$$s_1(P_m) \geq \cdots \geq s_m(P_m) > 0 > -s_m(P_m) \geq \cdots \geq -s_1(P_m).$$

Cauchy interlacing gives

$$\mu_m \geq s_m(P_m) \geq \mu_{m+1} \geq -s_m(P_m) \geq \mu_{m+2}.$$

If j is even, (5.4) and interlacing give

$$0 < \mu_{m+1} \leq s_m(P_m) \leq -\mu_{m+2};$$

if j is odd, they give

$$0 < -\mu_{m+1} \leq s_m(P_m) \leq \mu_m.$$

Every other eigenvalue is farther from zero. Hence μ_{m+1} attains the minimum modulus, with sign prescribed by (5.4); equality corresponds to multiplicity. At a spectral endpoint the zero is simple because A_n is similar to the Jacobi matrix A_n^{sym} , whose eigenvalues are simple. Thus the selected ordered branch extends continuously to zero at both ends of the closed gap. This proves the assertion. $\square$

5.2 Folded continuants

We use folded coordinates for the noncentral gaps and a separate interlacing argument for the parity-changing central gaps. Let U_j be the Chebyshev polynomial of the second kind, with $U_{-1} = 0$. Put

$$C_0 = x^2 + (1-a)^2 - s^2, \quad \zeta_s = \frac{1+a^2 + (1+a)s}{2a},$$

and define the folded discriminant and recurrence coefficient by

$$\Delta_{\text{fold}} = ((1+a-x)^2 - s^2)((1+a+x)^2 - s^2),$$

$$\mathfrak{b} = (1+a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3.$$

The global recurrence polynomial is

$$\begin{aligned} \Pi(z) &:= 1 - C_0 z + \mathfrak{b} z^2 - a^2 C_0 z^3 + a^4 z^4, \\ u, v \in \mathbb{R}, \quad 2a(u+v) &= C_0, \quad a^2(4uv+2) = \mathfrak{b}, \\ \implies \Pi(z) &= (1 - 2auz + a^2 z^2)(1 - 2avz + a^2 z^2). \end{aligned} \tag{5.9}$$

On a real sheet with $\Delta_{\text{fold}} \geq 0$, we additionally put

$$\xi_{\pm} = \frac{C_0 \pm \sqrt{\Delta_{\text{fold}}}}{4a}.$$

Then the pair $(u, v) = (\xi_+, \xi_-)$ satisfies the two symmetric relations in (5.9):

$$\xi_+ + \xi_- = \frac{C_0}{2a}, \quad \mathfrak{b} = a^2(4\xi_+\xi_- + 2).$$

We use the root presentation on these real sheets and the polynomial Π itself for the global transfer and generating identities. The five-minor transfer, characteristic polynomial, boundary residuals, and generating functions are derived in appendix B. Here we use their parity-unified consequence. For a differentiable real function f , write

$$[f]_{u,v} := \begin{cases} \frac{f(u) - f(v)}{u - v}, & u \neq v, \\ f'(u), & u = v. \end{cases}$$

Whenever the real pair u, v satisfies the two symmetric relations in (5.9), the divided-difference calculation in appendix C gives

$$q_{2m}(x, s) = a^m [\mathfrak{f}_m]_{u,v}, \quad \mathfrak{f}_m(t) = (t - \zeta_s)(U_m(t) + U_{m-1}(t)), \tag{5.10}$$

and

$$q_{2m+1}(x, s) = a^m [\mathfrak{g}_m]_{u,v}, \quad \mathfrak{g}_m(t) = \left((x-s)t - \frac{x(1+a^2) + 2as}{2a} \right) U_m(t). \tag{5.11}$$

The recurrence and generating identities are polynomial and hence global. The real-pair formulas apply when the two symmetric relations hold, with the displayed derivative as the collision value at $u = v$. When $\Delta_{\text{fold}} < 0$, the polynomial identities avoid choosing a complex square-root branch.

These formulas replace the singular-value problem by a signed scalar branch with parity already resolved. The next step identifies a geometric parametrization of that branch.

5.3 Half-angle chord parametrization

Put $h = -\frac{1}{2} \log a > 0$, so that $a = e^{-2h}$, and set

$$\mathfrak{s}_h = \sinh h, \quad \mathfrak{c}_h = \cosh h.$$

For $K \geq 2$, define

$$\begin{aligned} \Phi_K(\theta) &= \sqrt{\mathfrak{s}_h^2 + \sin^2 \theta} \frac{|\sin(K\theta)|}{\sin \theta}, \quad 0 \leq \theta \leq \pi, \\ \Psi_K(\eta) &= \sqrt{\mathfrak{s}_h^2 - \sinh^2 \eta} \frac{\sinh(K\eta)}{\sinh \eta}, \quad 0 \leq \eta \leq h. \end{aligned} \tag{5.12}$$

The quotients in (5.12) are interpreted continuously at $\theta = 0, \pi$ and $\eta = 0$. The following half-angle reduction also fixes the sign convention. Put

$$y = \cos \theta, \quad z = \cosh \eta, \quad \mathfrak{p}_\theta = \sqrt{\mathfrak{c}_h^2 - y^2}, \quad \mathfrak{p}_\eta = \sqrt{\mathfrak{c}_h^2 - z^2},$$

and let $L_0 = 2r/\mathfrak{c}_h$. If $x = L_0 y z$, $s = L_0 \mathfrak{p}_\theta \mathfrak{p}_\eta > 0$, and $\tau = \varsigma s$ with $\varsigma \in \{-1, 1\}$, then direct expansion gives

$$\begin{aligned} (1 + a + x)^2 - s^2 &= 4a(z + y)^2, & (1 + a - x)^2 - s^2 &= 4a(z - y)^2, \\ u = 2z^2 - 1, & v = 2y^2 - 1, & \zeta_\tau &= 2\mathfrak{c}_h^2 - 1 + 2\varsigma \mathfrak{p}_\theta \mathfrak{p}_\eta. \end{aligned} \quad (5.13)$$

Substitution of (5.13) shows that the real pair (u, v) satisfies the two symmetric relations in (5.9). For $K = 2m + 1$, use

$$U_m(2t^2 - 1) + U_{m-1}(2t^2 - 1) = U_{K-1}(t);$$

for $K = 2m + 2$, use

$$U_m(2t^2 - 1) = \frac{U_{K-1}(t)}{2t}.$$

Substitution of this pair in (5.10) and (5.11), respectively, reduces either equation $q_{K-1}(x, \tau) = 0$ to

$$\mathfrak{p}_\theta U_{K-1}(y) = \varsigma \mathfrak{p}_\eta U_{K-1}(z). \quad (5.14)$$

In the even-fold case the intermediate factorization is

$$(\mathfrak{p}_\eta + \varsigma \mathfrak{p}_\theta)(\mathfrak{p}_\eta U_{K-1}(z) - \varsigma \mathfrak{p}_\theta U_{K-1}(y)) = 0.$$

The first factor could vanish only if $\varsigma = -1$ and $\mathfrak{p}_\eta = \mathfrak{p}_\theta$, which would imply $y^2 = z^2$; the distinct-variable condition therefore leaves precisely (5.14). In the odd-fold case the reduction also contains the factor $\mathfrak{p}_\eta y + \varsigma \mathfrak{p}_\theta z$. On a distinct-variable sheet this factor is nonzero: if it vanished, squaring would give

$$(\mathfrak{c}_h^2 - z^2)y^2 = (\mathfrak{c}_h^2 - y^2)z^2,$$

and hence $y^2 = z^2$. Thus the factor is nonzero whenever the folded variables are distinct. On the outer lobe $U_{K-1}(z) > 0$, whereas, for $j\pi/K < \theta < (j+1)\pi/K$,

$$\operatorname{sgn}(U_{K-1}(\cos \theta)) = \operatorname{sgn}(\sin(K\theta)) = (-1)^j.$$

Moreover, $\mathfrak{p}_\theta |U_{K-1}(y)| = \Phi_K(\theta)$ and $\mathfrak{p}_\eta U_{K-1}(z) = \Psi_K(\eta)$. Thus (5.14) is equivalent to $\varsigma = (-1)^j$ and

$$\Phi_K(\theta) = \Psi_K(\eta), \quad (5.15)$$

with

$$s = \mathcal{S}_{\text{hyp}}(\theta, \eta) := \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 - \sinh^2 \eta)}, \quad x = \frac{2r}{\mathfrak{c}_h} \cos \theta \cosh \eta. \quad (5.16)$$

Consequently $q_{K-1}(x, (-1)^j s) = 0$. Conversely, for distinct real folded variables, their sum and product determine

$$x^2 = L_0^2 y^2 z^2, \quad s^2 = L_0^2 (\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2).$$

Thus, for distinct real variables and a fixed sign of x , the two chord relations are equivalent to the corresponding signed equation. The continuous divided-difference convention extends the determinant identities to a folded-variable collision; the selected middle branch is shown below to be collision-free. Here $K - 1$ is the matrix size. If the exterior argument has crossed through 1, write it as $z = \cos \phi$, $0 \leq \phi < \pi/K$. Then the chord equation is $\Phi_K(\theta) = \Phi_K(\phi)$, and the second factor in the formula for s is $\mathfrak{s}_h^2 + \sin^2 \phi$. We denote the resulting value by

$$\mathcal{S}_{\text{ell}}(\theta, \phi) = \frac{2r}{\mathfrak{c}_h} \sqrt{(\mathfrak{s}_h^2 + \sin^2 \theta)(\mathfrak{s}_h^2 + \sin^2 \phi)}.$$

5.4 Outer-lobe geometry and chord classification

To identify the chords that realize the middle branch of lemma 5.2, set

$$\mathcal{W}_K(u) = \sqrt{\mathfrak{c}_h^2 - u^2} |U_{K-1}(u)|, \quad |u| \leq \mathfrak{c}_h. \quad (5.17)$$

At $u = \pm \mathfrak{c}_h$, (5.17) denotes the continuous endpoint value 0. Then

$$\Phi_K(\theta) = \mathcal{W}_K(\cos \theta), \quad \Psi_K(\eta) = \mathcal{W}_K(\cosh \eta).$$

If $u_{\ell,K}^{(0)} = \cos \frac{\ell\pi}{K}$, then, away from the zeros,

$$\frac{d}{du} \log \mathcal{W}_K(u) = \sum_{\ell=1}^{K-1} \frac{1}{u - u_{\ell,K}^{(0)}} - \frac{u}{\mathfrak{c}_h^2 - u^2},$$

and its derivative is

$$-\sum_{\ell=1}^{K-1} \frac{1}{(u - u_{\ell,K}^{(0)})^2} - \frac{\mathfrak{c}_h^2 + u^2}{(\mathfrak{c}_h^2 - u^2)^2} < 0. \quad (5.18)$$

The strict negativity in (5.18) makes the logarithmic derivative strictly decreasing. At each finite nodal endpoint it tends to the appropriate signed infinity, and it tends to $-\infty$ as $u \uparrow \mathfrak{c}_h$. Thus there is exactly one lobe maximum in every nodal interval, including the outer interval $\mathcal{J}_K^{\text{out}} = (\cos \frac{\pi}{K}, \mathfrak{c}_h)$. Let u_K^{max} be the unique maximizer on $\mathcal{J}_K^{\text{out}}$. For every $0 \leq \ell < \mathcal{W}_K(u_K^{\text{max}})$, there is a unique solution

$$z_K(\ell) \in (u_K^{\text{max}}, \mathfrak{c}_h], \quad \mathcal{W}_K(z_K(\ell)) = \ell;$$

we call it the *endpoint-side outer solution*. If $z_K(\ell) \geq 1$, write it as $\cosh \eta$, $0 \leq \eta \leq h$; if it is at most 1, write it as $\cos \phi$, $0 \leq \phi < \pi/K$. These are two coordinates on the same outer lobe, joined at $z = 1$.

For a positive-half gap and $\theta \in (j\pi/K, (j+1)\pi/K)$, $\theta \leq \pi/2$, a *j-chord* is the pair

$$y = \cos \theta, \quad z = z_K(\mathcal{W}_K(y)), \quad \mathcal{W}_K(y) = \mathcal{W}_K(z),$$

together with the coordinate-free values

$$x = L_0 y z, \quad s = L_0 \sqrt{(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)};$$

its signed root is $(-1)^j s$. The term endpoint-side always refers to the part of the outer lobe adjacent to $\mathfrak{c}_h$, independently of whether it is described by the hyperbolic or elliptic coordinate.

Reflection reduces the branch classification to a positive-half gap, so $j \geq 1$ and $\theta \leq \pi/2$. Put $\phi_{\text{rem}} = \theta - j\pi/K \in (0, \pi/K)$. Since $|\sin(K\phi_{\text{rem}})| = |\sin(K\theta)|$ and

$$\mathcal{A}_h(t) = \frac{\sqrt{\mathfrak{s}_h^2 + \sin^2 t}}{\sin t}$$

is strictly decreasing on $(0, \pi/2]$,

$$\Phi_K(\theta) < \Phi_K(\phi_{\text{rem}}) \leq \max_{u \in \mathcal{J}_K^{\text{out}}} \mathcal{W}_K(u). \quad (5.19)$$

Thus the endpoint-side outer solution exists for every inner parameter.

At $x = \lambda_{K-1,k} = 2r \cos \frac{k\pi}{K}$, $s = 0$, the two folded variables are $\cos \frac{2k\pi}{K}$ and $\cosh(2h)$, or, in half variables, $\cos \frac{k\pi}{K}$ and $\mathfrak{c}_h$. Moreover, zero is a simple eigenvalue of $B_{K-1}(\lambda_{K-1,k})$: the matrix $\lambda_{K-1,k} I_{K-1} - A_{K-1}$ has rank $K-2$, and J_{K-1} is invertible. The implicit-function theorem therefore gives, near either gap endpoint, a real continuation of the middle signed root and real half variables $y = \cos \theta$ and z .

5.5 Exhaustion of the selected chord sheet

Lemma 5.3 (Endpoint-side chord-sheet exhaustion). *Let $K \geq 3$ and let $j \geq 1$ be an integer. Under the conventions of Equations (5.15) and (5.16), if*

$$\frac{(j+1)\pi}{K} \leq \frac{\pi}{2},$$

then every point of the ordered middle branch in the corresponding open positive gap has a selected endpoint-side j -chord representation with $j\pi/K < \theta < (j+1)\pi/K$, and every θ in this interval gives a point of that branch. The signed root is $(-1)^j s$, its magnitude is $s_{\min}(xI_{K-1} - A_{K-1})$, and the two folded variables are distinct. This includes a gap adjacent to zero. Separately, if K is odd and $j = (K-1)/2$, the same bidirectional statement holds from the positive spectral endpoint to the center of the single central gap, with $j\pi/K < \theta \leq \pi/2$.

The proof is deferred to appendix D; the lemma identifies every middle-branch value used below with the selected chord, and hence with the least singular value.

5.6 Noncentral gap comparison

Lemma 5.4 (Noncentral gap comparison). *Let $j \geq 1$ and $N \geq 2(j+1)$ be integers, and put $M = N+1$. For every $x \in G_{M-1,j}^\circ$, there is $\hat{x} \in G_{N-1,j}^\circ$ such that*

$$g_{N-1}(\hat{x}) > g_{M-1}(x).$$

Proof. Fix $x \in G_{M-1,j}^\circ$. By lemma 5.3, its middle branch has a hyperbolic or elliptic chord representation with

$$\theta \in \left(\frac{j\pi}{M}, \frac{(j+1)\pi}{M} \right).$$

Set $\kappa = N/M$ and $\theta_0 = \theta/\kappa$. The hypotheses place θ_0 in the corresponding N -nodal interval, with every trigonometric argument below in $(0, \pi/2]$.

First suppose $\Phi_M(\theta) = \Psi_M(\eta)$. Concavity of sine gives

$$\frac{\Phi_N(\theta_0)}{\Phi_M(\theta)} = \frac{\mathcal{A}_h(\theta_0)}{\mathcal{A}_h(\theta)} > \kappa. \quad (5.20)$$

Indeed, $\theta = \kappa\theta_0$ and strict concavity gives $\sin \theta > \kappa \sin \theta_0$; hence

$$\mathcal{A}_h(\theta_0)^2 - \kappa^2 \mathcal{A}_h(\theta)^2 = (1 - \kappa^2) + \mathfrak{s}_h^2 \left(\frac{1}{\sin^2 \theta_0} - \frac{\kappa^2}{\sin^2 \theta} \right) > 0.$$

On the other hand,

$$\frac{\Psi_N(\eta)}{\Psi_M(\eta)} = \frac{\sinh(N\eta)}{\sinh(M\eta)} \leq \kappa, \quad (5.21)$$

with strict inequality for $\eta > 0$, by convexity of $\sinh$ on the positive half-axis. Together, (5.20) and (5.21) imply $\Phi_N(\theta_0) > \Psi_N(\eta)$. Since Φ_N vanishes at the right endpoint $(j+1)\pi/N$, its rightmost solution $\hat{\theta}$ of

$$\Phi_N(\hat{\theta}) = \Psi_N(\eta)$$

exists by the intermediate value theorem and satisfies $\hat{\theta} > \theta_0 > \theta$. Strict log-concavity of $\mathcal{W}_N$ makes it the unique solution on the descending half of that lobe. The same exterior variable is endpoint-side for the N -problem as well. Indeed, for $u = \cosh \eta > 1$,

$$\frac{d}{du} \log \mathcal{W}_M(u) - \frac{d}{du} \log \mathcal{W}_N(u) = \frac{M \coth(M\eta) - N \coth(N\eta)}{\sinh \eta} > 0.$$

The numerator is positive because, for $\eta > 0$,

$$\frac{d}{dt}(t \coth(t\eta)) = \frac{\sinh(t\eta) \cosh(t\eta) - t\eta}{\sinh^2(t\eta)} > 0.$$

The endpoint-side branch is characterized by a nonpositive logarithmic derivative. The displayed difference therefore makes the N -derivative strictly negative. The case $\eta = 0$ follows by continuity. The chord classification above says that the constructed value $\mathcal{S}_{\text{hyp}}(\widehat{\theta}, \eta)$ is the least singular value at its constructed abscissa. Formula (5.16) and the strict increase of sine on the interval now give

$$\mathcal{S}_{\text{hyp}}(\widehat{\theta}, \eta) > \mathcal{S}_{\text{hyp}}(\theta, \eta). \quad (5.22)$$

Suppose next that the exterior endpoint is elliptic, so that $\Phi_M(\theta) = \Phi_M(\phi)$, $0 < \phi < \pi/M$. (The case $\phi = 0$ is the limiting hyperbolic case already covered.) Put

$$\theta_0 = \theta/\kappa, \quad \phi_0 = \phi/\kappa.$$

The ratio

$$\mathcal{A}_{N,M}^\Phi(t) = \frac{\Phi_N(t/\kappa)}{\Phi_M(t)} = \frac{\mathcal{A}_h(t/\kappa)}{\mathcal{A}_h(t)}$$

is strictly increasing while $0 < t < t/\kappa \leq \pi/2$. Indeed,

$$\frac{d}{dt} \log \mathcal{A}_{N,M}^\Phi(t) = \mathfrak{d}_h(t) - \kappa^{-1} \mathfrak{d}_h(t/\kappa) > 0, \quad \mathfrak{d}_h(t) = \frac{\mathfrak{s}_h^2 \cot t}{\mathfrak{s}_h^2 + \sin^2 t},$$

because

$$t \mathfrak{d}_h(t) = (t \cot t) \frac{\mathfrak{s}_h^2}{\mathfrak{s}_h^2 + \sin^2 t}$$

is strictly decreasing: both positive factors decrease on $(0, \pi/2)$, and the second decreases strictly. Since $\theta > \pi/M > \phi$, it follows that $\Phi_N(\theta_0) > \Phi_N(\phi_0)$. The rightmost solution of

$$\Phi_N(\widehat{\theta}) = \Phi_N(\phi_0)$$

therefore has $\widehat{\theta} > \theta_0$. Its exterior variable is endpoint-side. Indeed, if $\Lambda_K(t) = (\log \Phi_K)'(t)$, then on the endpoint-side part of the first outer lobe $\Lambda_M(\phi) \geq 0$, and differentiation of $\Phi_N(t/\kappa) = \mathcal{A}_{N,M}^\Phi(t) \Phi_M(t)$ gives

$$\kappa^{-1} \Lambda_N(t/\kappa) = \Lambda_M(t) + \frac{(\mathcal{A}_{N,M}^\Phi)'(t)}{\mathcal{A}_{N,M}^\Phi(t)} > 0.$$

Because $\Lambda_K(\phi) = -\sin \phi (\log \mathcal{W}_K)'(\cos \phi)$, positivity of $\Lambda_N(\phi_0)$ is exactly the endpoint-side condition for the N -outer solution. The one-maximum property now fixes the label, and the chord classification says again that $\mathcal{S}_{\text{ell}}(\widehat{\theta}, \phi_0)$ is the least singular value at the constructed abscissa. Here $\widehat{\theta} > \theta_0 > \theta$ and $\phi_0 > \phi$, with all four angles below $\pi/2$. Thus both sine factors in the elliptic version of (5.16) have increased, at least one strictly, and hence

$$\mathcal{S}_{\text{ell}}(\widehat{\theta}, \phi_0) > \mathcal{S}_{\text{ell}}(\theta, \phi). \quad (5.23)$$

Equations (5.22) and (5.23), together with the coordinate formula for x , place the constructed abscissa $\widehat{x}$ in $G_{N-1,J}^\circ$. Chord exhaustion identifies the original and constructed heights with $g_{M-1}(x)$ and $g_{N-1}(\widehat{x})$, respectively, proving the claim. $\square$

5.7 Central gap comparison

For $m \geq 1$, define the even-order central height by

$$\mathfrak{h}_m^e = s_{\min}(aI_m + V_m). \quad (5.24)$$

By reflection, the two gaps adjacent to the zero eigenvalue of A_{2m+1} have the same height; denote it by

$$\mathfrak{h}_m^o = \max_{0 \leq x \leq 2r \sin \frac{\pi}{2m+2}} s_{\min}(xI_{2m+1} - A_{2m+1}). \quad (5.25)$$

The central heights defined in (5.24) and (5.25) satisfy the following strict alternation.

Lemma 5.5 (Central gap comparison). *For every $m \geq 1$,*

$$\mathfrak{h}_m^e > \mathfrak{h}_m^o > \mathfrak{h}_{m+1}^e. \quad (5.26)$$

Proof. Put

$$\zeta = \frac{a + a^{-1}}{2}.$$

For every $j \geq 1$, there is a unique $\rho_j > \cos \frac{\pi}{j+1}$ such that

$$aU_j(\rho_j) = U_{j-1}(\rho_j). \quad (5.27)$$

Indeed, the Christoffel–Darboux identity gives, above the largest zero of U_j ,

$$\frac{d}{dt} \frac{U_{j-1}(t)}{U_j(t)} = -\frac{2 \sum_{k=0}^{j-1} U_k(t)^2}{U_j(t)^2} < 0.$$

The ratio decreases from $+\infty$ to 0, proving existence and uniqueness. A three-term determinant recurrence gives

$$\det \left((aI_j + V_j)^*(aI_j + V_j) - (1 + a^2 - 2at)I_j \right) = a^j \left(U_j(t) - a^{-1}U_{j-1}(t) \right). \quad (5.28)$$

Thus the largest root ρ_j gives the least eigenvalue. Together with

$$U_j(t)^2 + U_{j-1}(t)^2 - 2tU_j(t)U_{j-1}(t) = 1,$$

equations (5.27) and (5.28) give

$$(\mathfrak{h}_j^e)^2 = 1 + a^2 - 2a\rho_j = \frac{1}{U_j(\rho_j)^2}, \quad \zeta - \rho_j = \frac{1}{2U_{j-1}(\rho_j)U_j(\rho_j)}. \quad (5.29)$$

We first identify $\mathfrak{h}_m^e$ as the central-gap height of A_{2m} . Let

$$P_{oe} = [\mathbf{e}_1, \mathbf{e}_3, \dots, \mathbf{e}_{2m-1}, \mathbf{e}_2, \mathbf{e}_4, \dots, \mathbf{e}_{2m}]$$

be the permutation matrix that places odd coordinates first. Then

$$P_{oe}^T A_{2m} P_{oe} = \begin{bmatrix} 0 & I_m + aV_m^T \\ aI_m + V_m & 0 \end{bmatrix}.$$

Moreover,

$$(I_m + aV_m^T)^*(I_m + aV_m^T) - J_m(aI_m + V_m)^*(aI_m + V_m)J_m = (1 - a^2)\mathbf{e}_m\mathbf{e}_m^T \geq 0.$$

The singular values of the block off-diagonal matrix are those of its two off-diagonal blocks. The Gram inequality shows that $s_{\min}(I_m + aV_m^T) \geq s_{\min}(aI_m + V_m) = \mathfrak{h}_m^e$, and hence $s_{\min}(A_{2m}) = \mathfrak{h}_m^e$. To maximize over the central gap, put $K = 2m + 1$ and use the half variables

$$y = \cos \theta, \quad z = \cosh \eta \quad \text{or} \quad z = \cos \phi.$$

On the positive half of the central middle branch, the preceding chord classification gives

$$\mathcal{W}_K(y) = \mathcal{W}_K(z), \quad x = \frac{2r}{\mathfrak{c}_h} yz, \quad s^2 = \frac{4a}{\mathfrak{c}_h^2} (\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2), \quad (5.30)$$

where $0 \leq y < \cos \frac{m\pi}{K}$ is in the central lobe and z is the endpoint-side outer solution. The function $\mathcal{W}_K$ is even, and strict log-concavity on the central lobe gives

$$\mathcal{W}_K(y) < \mathcal{W}_K(0) = \mathfrak{c}_h \quad (y > 0).$$

Let

$$z_0 = \sqrt{\frac{1 + \rho_m}{2}}.$$

The half-angle identity and (5.27) give

$$U_{2m}(z_0) = U_m(\rho_m) + U_{m-1}(\rho_m) = (1 + a)U_m(\rho_m).$$

Together with (5.29), this yields

$$\mathcal{W}_K(z_0) = \mathfrak{c}_h, \quad 4a(\mathfrak{c}_h^2 - z_0^2) = 2a(\zeta - \rho_m) = (\mathfrak{h}_m^e)^2. \quad (5.31)$$

At $x = 0$, lemma 5.2 and chord exhaustion identify the nearest root with $\mathfrak{h}_m^e$, so z_0 is the endpoint-side outer solution of level $\mathfrak{c}_h$. If $x \neq 0$, the level in (5.30) is strictly below $\mathfrak{c}_h$, and its endpoint-side solution satisfies $z > z_0$. Equations (5.30)–(5.31) then give

$$s^2 < 4a(\mathfrak{c}_h^2 - z_0^2) = (\mathfrak{h}_m^e)^2.$$

Reflection gives the same strict bound on the other half of the gap. Therefore the central-gap height is $\mathfrak{h}_m^e$, uniquely attained at $x = 0$.

Set

$$\omega_m^e = \cos \frac{\pi}{m+1}, \quad \mathcal{V}_m(t) = \sqrt{(\zeta - t)(1 + t)} U_m(t).$$

For the central specialization, take $K = 2m + 2$ and set

$$t_{\text{in}} = 2y^2 - 1 = \cos(2\theta), \quad t_{\text{out}} = 2z^2 - 1.$$

On the positive central gap, $-1 < t_{\text{in}} < -\omega_m^e$, while the endpoint-side outer variable satisfies $\omega_m^e < t_{\text{out}} < \zeta$, where $\zeta = 2\mathfrak{c}_h^2 - 1$. The identity $U_{2m+1}(t) = 2tU_m(2t^2 - 1)$ gives

$$\begin{aligned} \mathcal{W}_{2m+2}(y)^2 &= (\zeta - t_{\text{in}})(1 + t_{\text{in}})U_m(t_{\text{in}})^2, \\ \mathcal{W}_{2m+2}(z)^2 &= (\zeta - t_{\text{out}})(1 + t_{\text{out}})U_m(t_{\text{out}})^2. \end{aligned}$$

Because $\zeta + 1 = 2\mathfrak{c}_h^2$, the formulas $x = (2r/\mathfrak{c}_h)yz$ and $s^2 = (4a/\mathfrak{c}_h^2)(\mathfrak{c}_h^2 - y^2)(\mathfrak{c}_h^2 - z^2)$ become the following chord parametrization:

$$\begin{aligned} -1 < t_{\text{in}} < -\omega_m^e, \quad \omega_m^e < t_{\text{out}} < \zeta, \quad \mathcal{V}_m(t_{\text{out}}) &= |\mathcal{V}_m(t_{\text{in}})|, \\ x^2 &= \frac{2a(1 + t_{\text{out}})(1 + t_{\text{in}})}{\zeta + 1}, \quad s^2 = \frac{2a(\zeta - t_{\text{out}})(\zeta - t_{\text{in}})}{\zeta + 1}, \end{aligned} \quad (5.32)$$

where t_{out} is the descending, or largest, solution. To derive (5.32) from (5.15), use

$$\mathfrak{s}_h^2 + \sin^2 \theta = \frac{\zeta - t_{\text{in}}}{2}, \quad \mathfrak{s}_h^2 - \sinh^2 \eta = \frac{\zeta - t_{\text{out}}}{2}.$$

Continuation through $t_{\text{out}} = 1$ uses the elliptic coordinate $z = \cos \phi$.

Let $\rho = \rho_m$. The function $\log \mathcal{V}_m$ is strictly concave on (ω_m^e, ζ) . If $(\log \mathcal{V}_m)'(\rho) \geq 0$, then ρ lies at or before the unique lobe maximum. The descending root lies at or after that maximum, so $t_{\text{out}} \geq \rho$. Now suppose $(\log \mathcal{V}_m)'(\rho) < 0$. Then $\rho > a$: if $\rho \leq a < 1$, then $\rho < 1$, hence $\rho^2 - 1 \neq 0$, and the following quotient is well defined:

$$\frac{U'_m(\rho)}{U_m(\rho)} = \frac{m\rho - (m+1)a}{\rho^2 - 1}$$

would give

$$\begin{aligned} (\log \mathcal{V}_m)'(\rho) &= \frac{(2m+2)a + 1 - (2m+1)\rho}{2(1-\rho^2)} - \frac{1}{2(\zeta - \rho)} \\ &\geq \frac{1}{2(1-\rho)} - \frac{1}{2(\zeta - \rho)} \geq 0, \end{aligned}$$

which contradicts the assumed negative derivative. Hence $\rho > a$. Writing $t_{\text{in}} = -\cos \phi_c$, $0 < \phi_c < \pi/(m+1)$, gives

$$\begin{aligned} |\mathcal{V}_m(t_{\text{in}})|^2 &= \frac{(\zeta + \cos \phi_c) \sin^2((m+1)\phi_c)}{1 + \cos \phi_c} \\ &< \frac{\zeta + \omega_m^e}{1 + \omega_m^e} < \frac{1+a}{2a} < \frac{1+\rho}{2a} = \mathcal{V}_m(\rho)^2. \end{aligned}$$

Now ρ is on the descending side and $|\mathcal{V}_m(t_{\text{in}})| < \mathcal{V}_m(\rho)$, so the unique descending root again satisfies $t_{\text{out}} > \rho$. In either case, $t_{\text{out}} \geq \rho$, and $t_{\text{in}} > -1$ makes the following inequality strict:

$$s^2 < 2a(\zeta - \rho) = (\mathfrak{h}_m^e)^2.$$

This holds along the whole central sheet, so $\mathfrak{h}_m^o < \mathfrak{h}_m^e$. Thus the odd-order central height lies strictly below the preceding even-order central height. We next compare it with the following even-order height.

Lower comparison $\mathfrak{h}_m^o > \mathfrak{h}_{m+1}^e$: scalar estimate

Put $L = m+1 \geq 2$, and abbreviate

$$\begin{aligned} \rho &= \rho_L, \quad u_- = U_{L-1}(\rho), \quad u_+ = U_L(\rho), \quad \omega_L = \cos \frac{\pi}{2L}, \quad \Delta_\omega = 1 - \omega_L, \\ \mathfrak{D}_L &= 1 + 2(\rho + \omega_L)u_-u_+, \quad \xi = \rho - \frac{\Delta_\omega}{\mathfrak{D}_L}. \end{aligned} \tag{5.33}$$

Thus $a = u_-/u_+$ and $\mathfrak{h}_L^e = 1/u_+$.

The required inequality is

$$(1 + \omega_L)(1 + \xi)^2 U_{L-1}(\xi)^2 < (\rho + \omega_L) \mathfrak{D}_L. \tag{5.34}$$

Appendix E proves (5.34) for every $L \geq 2$ from the Chebyshev identity and strict log-concavity, treating the three cases $\rho > 1$, $\rho = 1$, and $\rho < 1$.

Certification of the least singular value

Since $a > 0$, $\Delta_\omega > 0$, $\rho + \omega_L > 0$, and $\zeta + \omega_L > 0$, let $x_* > 0$ be the positive square root determined by

$$x_*^2 = \frac{2a\Delta_\omega(\rho + \omega_L)}{\zeta + \omega_L}, \quad \tau_* = \mathfrak{h}_L^e. \tag{5.35}$$

The algebraic certification of the choice (5.35) is given in appendix F. Substitution of the distinct folded pair $u = \xi$, $v = -\omega_L$ into (5.11), followed by (5.34), proves

$$q_{2L-1}(x_*, \tau_*)q_{2L-1}(x_*, -\tau_*) > 0. \quad (5.36)$$

At $x = 0$, the second-smallest singular value of A_{2L-1} is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

The same appendix proves the strict separation

$$x_* + \tau_* < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}. \quad (5.37)$$

The product sign and second-singular-value separation identify the relevant signed root.

Let $\mu_1, \dots, \mu_{2L-1}$ be the eigenvalues of the real symmetric matrix

$$B_{2L-1}(x_*) = (x_* I_{2L-1} - A_{2L-1})J_{2L-1}.$$

Right multiplication by J_{2L-1} preserves singular values, so the numbers $|\mu_i|$ are precisely the singular values of $x_* I_{2L-1} - A_{2L-1}$. Moreover,

$$q_{2L-1}(x_*, \tau_*)q_{2L-1}(x_*, -\tau_*) = \prod_{i=1}^{2L-1} (\mu_i^2 - \tau_*^2).$$

Positivity of the product in (5.36) implies that the number of indices with $|\mu_i| < \tau_*$ is even. On the other hand, Weyl's singular-value perturbation inequality and (5.37) give, for the second-smallest singular value s_{next} ,

$$s_{\text{next}}(x_* I_{2L-1} - A_{2L-1}) \geq s_{L,2}^{(0)} - x_* > \tau_*,$$

so at most one index can satisfy $|\mu_i| < \tau_*$. Parity therefore gives $|\mu_i| \geq \tau_*$ for every i , and strict positivity of the product rules out equality. Hence $s_{\min}(x_* I_{2L-1} - A_{2L-1}) > \mathfrak{h}_L^e$. Since $0 < x_* < 2r \sin \frac{\pi}{2L}$, the point lies strictly in the positive central gap, and $\mathfrak{h}_{L-1}^o > \mathfrak{h}_L^e$. Together with the upper comparison $\mathfrak{h}_m^e > \mathfrak{h}_m^o$, this proves (5.26). $\square$

5.8 Assembly of the global comparison

Proof of Lemma 5.1. Reflection and the central interlacing chain now assemble the local comparisons into the global barrier inequality. Put $N = n + 1$ and $M = N + 1$, so the larger and smaller matrices have orders $M - 1$ and $N - 1$, respectively. If $\Sigma = \text{diag}(1, -1, 1, -1, \dots)$, then $\Sigma A_n \Sigma = -A_n$, so reflection reduces every noncentral comparison to a positive gap. For each positive noncentral gap $G_{M-1,j}$, choose $x_j \in G_{M-1,j}^\circ$ attaining $\gamma_{M-1,j}$. Then $j \geq 1$ and $(j + 1)\pi/N \leq \pi/2$, so lemma 5.4 produces a point $\widehat{x}_j \in G_{N-1,j}^\circ$, and chord exhaustion gives

$$\gamma_{N-1,j} \geq g_{N-1}(\widehat{x}_j) > g_{M-1}(x_j) = \gamma_{M-1,j}.$$

Reflection gives the same inequality on the corresponding reflected gap.

By lemma 5.5, if $M = 2\ell_* + 1$, the remaining single central height of the larger, even-order matrix is $\mathfrak{h}_{\ell_*}^e$, whereas each central height of the smaller odd-order matrix is $\mathfrak{h}_{\ell_*-1}^o > \mathfrak{h}_{\ell_*}^e$. If $M = 2\ell_* + 2$, the two central heights of the larger odd-order matrix equal $\mathfrak{h}_{\ell_*}^o < \mathfrak{h}_{\ell_*}^e$, where $\mathfrak{h}_{\ell_*}^e$ is the single central height of the smaller even-order matrix. Thus every height of the larger problem is strictly smaller than at least one height of the smaller problem. By (5.1), choose j_* with $\gamma_{M-1,j_*} = \max_j \gamma_{M-1,j}$. Then

$$\gamma_{M-1} = \gamma_{M-1,j_*} < \max_k \gamma_{N-1,k} = \gamma_{N-1}.$$

Since $M - 1 = n + 1$ and $N - 1 = n$, this proves $\gamma_{n+1} < \gamma_n$. $\square$

6 Quantitative bounds and the transition scale

The connectedness criterion (4.2) and the strict comparison of lemma 5.1 reduce the remaining problem to estimating γ_n . Put $N = n + 1$ and recall $\mathcal{L}_n(a)$ and $\mathcal{U}_n(a)$ from (3.8)–(3.9).

Proposition 6.1 (Optimal-order gap bounds). *For every $n \geq 2$,*

$$\mathcal{L}_n(a) \leq \gamma_n \leq \mathcal{U}_n(a). \quad (6.1)$$

In particular, both bounds have order $(n + 1)a^{n/2}$, with constants depending only on a .

Proof. We prove the two inequalities in (6.1) separately. For $x = 2r \cos \theta \in I_n^{\text{sp}}$, set

$$\mathbf{v} = (v_j)_{j=1}^n, \quad v_j = r^{j-1} \sin(j\theta), \quad 1 \leq j \leq n.$$

The residual $(xI_n - A_n)\mathbf{v}$ is supported in its last coordinate, where it equals $r^n \sin(N\theta)$. Thus

$$g_n(2r \cos \theta) \leq \frac{r^n |\sin(N\theta)|}{\left(\sum_{j=1}^n r^{2j-2} \sin^2(j\theta)\right)^{1/2}} \leq r^n \frac{|\sin(N\theta)|}{\sin \theta}.$$

Since $\pi/N \leq \theta \leq (N - 1)\pi/N$, this proves the upper bound.

For the lower bound, put $p_\theta = re^{i\theta}$, $q_\theta = re^{-i\theta}$, so that $p_\theta + q_\theta = x$ and $p_\theta q_\theta = a$. For $\mathbf{u} = (u_j)_{j=1}^n \in \mathbb{C}^n$, put $u_0 = u_{n+1} = 0$, and define maps from $\mathbb{C}^n$ to $\mathbb{C}^{n+1}$ by

$$(\mathbf{C}_\theta \mathbf{u})_j = p_\theta u_j - u_{j+1}, \quad (\mathbf{D}_\theta \mathbf{u})_j = u_j - \overline{q_\theta} u_{j+1}, \quad 0 \leq j \leq n.$$

Direct multiplication, using $p_\theta + q_\theta = x$ and $p_\theta q_\theta = a$, gives

$$xI_n - A_n = \mathbf{D}_\theta^* \mathbf{C}_\theta. \quad (6.2)$$

The two Gram matrices associated with the factorization (6.2) are diagonally unitarily similar to the tridiagonal Toeplitz matrix with diagonal $1 + r^2$ and off-diagonal $-r$. Its eigenvalues are $1 + r^2 - 2r \cos \frac{k\pi}{N}$, $1 \leq k \leq n$. Consequently, for every $\mathbf{u} \in \mathbb{C}^n$,

$$\begin{aligned} \|\mathbf{C}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2, \\ \|\mathbf{D}_\theta \mathbf{u}\|_2 &\geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right)^{1/2} \|\mathbf{u}\|_2. \end{aligned} \quad (6.3)$$

Put

$$S_N = \sum_{j=0}^n r^{2j} = \frac{1 - r^{2N}}{1 - r^2}.$$

Unit normal vectors to the two hyperplane ranges are

$$\mathbf{u}_C = S_N^{-1/2} (\overline{p_\theta}^n, \overline{p_\theta}^{n-1}, \dots, 1)^T, \quad \mathbf{u}_D = S_N^{-1/2} (1, q_\theta, \dots, q_\theta^n)^T.$$

In the polar decompositions

$$\mathbf{C}_\theta = Q_C (\mathbf{C}_\theta^* \mathbf{C}_\theta)^{1/2}, \quad \mathbf{D}_\theta = Q_D (\mathbf{D}_\theta^* \mathbf{D}_\theta)^{1/2},$$

the matrices Q_C, Q_D are isometries onto these ranges. Two hyperplanes in $\mathbb{C}^{n+1}$ intersect in dimension $n - 1$, so the cross-Gram matrix $Q_D^* Q_C$ has $n - 1$ singular values equal to 1; its final singular value is $|\langle \mathbf{u}_D, \mathbf{u}_C \rangle|$, the cosine of the nontrivial principal angle. The geometric sum gives

$$|\langle \mathbf{u}_D, \mathbf{u}_C \rangle| = \frac{r^n}{S_N} \left| \sum_{j=0}^{N-1} e^{i(2j - (N-1))\theta} \right|$$

$$= \frac{r^n(1-r^2)}{1-r^{2N}} \left| \frac{\sin(N\theta)}{\sin\theta} \right| =: \chi_N(\theta). \quad (6.4)$$

Finally,

$$xI_n - A_n = (D_\theta^* D_\theta)^{1/2} Q_D^* Q_C (C_\theta^* C_\theta)^{1/2}.$$

Using $s_{\min}(XYZ) \geq s_{\min}(X)s_{\min}(Y)s_{\min}(Z)$ and (6.3), with χ_N as defined in (6.4), gives

$$g_n(2r \cos \theta) \geq \left(1 + r^2 - 2r \cos \frac{\pi}{N}\right) \chi_N(\theta).$$

Putting $\theta = 3\pi/(2N)$, which lies in the first spectral gap, proves the lower bound.

Finally,

$$\mathcal{U}_n(a) \leq \frac{1}{2}(n+1)r^n,$$

while

$$\mathcal{L}_n(a) \geq C_{\text{lb}}(a)(n+1)r^n, \quad C_{\text{lb}}(a) = \frac{2(1-r)^2(1-r^2)}{3\pi}. \quad (6.5)$$

The estimate (6.5) and the preceding upper estimate follow from $\sin \frac{\pi}{N} \geq 2/N$, $1+r^2-2r \cos \frac{\pi}{N} \geq (1-r)^2$, and $\sin \frac{3\pi}{2N} \leq 3\pi/(2N)$. This proves the asserted optimal order up to constants depending only on a . $\square$

Proof of Theorem 3.3. By Proposition 6.1, $\gamma_n \leq \mathcal{U}_n(a) \rightarrow 0$. Lemma 5.1 says that γ_n is strictly decreasing. Equation (4.2) therefore shows that, once the pseudospectrum is connected, it stays connected at every larger matrix order, which proves the set identity in part 3. Contractibility and the connectedness criterion are exactly (4.2).

If $\mathcal{L}_n(a) \geq \varepsilon$, then $\gamma_n \geq \varepsilon$, so the pseudospectrum at order n is disconnected. If $\mathcal{U}_n(a) < \varepsilon$, then $\gamma_n < \varepsilon$, so it is connected. Strict decrease of γ_n now gives (3.10).

For $t > 0$, define

$$\mathfrak{e}_n = (n+1)r^n, \quad \mathcal{T}_r(t) = \min\{N_0 \geq 2 : \mathfrak{e}_m < t \text{ for every } m \geq N_0\}.$$

This minimum exists because $\mathfrak{e}_m \rightarrow 0$. Because persistence gives

$$N_c(a, \varepsilon) = \min\{N \geq 2 : \gamma_m < \varepsilon \text{ for every } m \geq N\},$$

the bounds $C_{\text{lb}}(a)\mathfrak{e}_m \leq \gamma_m \leq \mathfrak{e}_m/2$ give

$$\mathcal{T}_r(\varepsilon/C_{\text{lb}}(a)) \leq N_c(a, \varepsilon) \leq \mathcal{T}_r(2\varepsilon). \quad (6.6)$$

Elementary inversion of the envelope threshold

Let

$$\beta = -\log r = \frac{1}{2} \log(1/a), \quad L(t) = \log(1/t), \quad S_r(t) = \frac{L(t) + \log L(t)}{\beta}.$$

For every fixed $D \in \mathbb{R}$, direct substitution gives

$$\beta(S_r(t) + D) - \log(S_r(t) + D + 1) - L(t) = \beta D + \log \beta + o(1) \quad (t \downarrow 0).$$

Indeed, $\beta S_r = L + \log L$ and $\log(S_r + D + 1) = \log L - \log \beta + o(1)$. Choose constants $D_- < -(\log \beta)/\beta < D_+$. For all sufficiently small t , the preceding residual is negative at $S_r(t) + D_-$ and positive at $S_r(t) + D_+$. Since $S_r(t) \rightarrow \infty$ and

$$\frac{d}{dx}((x+1)e^{-\beta x}) = e^{-\beta x}(1 - \beta(x+1)),$$

this function is decreasing on $[S_r(t) + D_-, \infty)$ for all sufficiently small t . The strict integer tail condition therefore gives

$$\lfloor S_r(t) + D_- \rfloor < \mathcal{T}_r(t) \leq \lceil S_r(t) + D_+ \rceil.$$

Thus $\mathcal{T}_r(t) = S_r(t) + O_r(1)$. Replacing t by a fixed positive multiple changes $S_r(t)$ by only $O_r(1)$; applying this to the two sides of (6.6) proves (3.11).

Exact two-site endpoint

For $|x| \leq r$,

$$\det((xI_2 - A_2)^*(xI_2 - A_2) - a^2 I_2) = x^2(x^2 - 2a - 2a^2) \leq 0.$$

The matrix inside the determinant is Hermitian, and the nonpositive product of its two eigenvalues forces its least eigenvalue to be nonpositive. Thus $s_{\min}(xI_2 - A_2) \leq a$. At $x = 0$, the singular values of $-A_2$ are 1 and a , so equality is attained and $\gamma_2 = a$. Hence (4.2) gives $N_c(a, \varepsilon) = 2$ if and only if $\varepsilon > a$. This proves all assertions. $\square$

7 Numerical illustrations

Figures 1 and 2 illustrate two aspects of the matrix-order transition: coalescence of pseudospectral components and decay of the real-axis backward-error barrier.

At each sampled point, the least singular value was computed by a double-precision singular value decomposition. The contour plots use a 561-by-361 Cartesian grid. Each spectral gap was scanned at 121 equally spaced points before bounded scalar optimization was used to estimate γ_n . The displayed component counts were also evaluated on 1001 points per gap, with the 1-Lipschitz dependence of the least singular value controlling variation between adjacent sample points.

8 Discussion

The boundary regimes are singular limits of the nonnormal interior: the threshold scale changes from $c(n+1)a^{n/2}$ for $0 < a < 1$ to $\pi c/(n+1)$ at $a = 1$ and zero at $a = 0$. Accordingly, the remainder $O_a(1)$ in (3.11) is pointwise in fixed a and loses uniformity near the boundaries.

For every irreducible family, the connected orders form an integer tail with initial index $N_T(\alpha, d, \beta; \varepsilon)$; at the reducible boundary, the tail begins at $n = 2$.

The vertical-contraction theorem separates the transferable topology from the Toeplitz-specific algebra. For another matrix family with spectrum on an affine line, contraction toward that line yields component contractibility and a one-dimensional connectedness criterion. Adjacent-order comparison then becomes a problem for the resulting barrier; in the present family it is resolved by folded continuants and chord geometry.

We use the Euclidean vector norm and the corresponding full ball of complex perturbations in the induced operator norm. Structured variants generally have different connectedness thresholds; see [11] for Toeplitz-structured pseudospectra. Block Toeplitz matrices and periodic k -Toeplitz families are natural settings for extending the order comparison.

Restricting the nearest-neighbor Hatano–Nelson hopping law [14, 15] to an open finite chain and, if necessary, reversing its orientation gives, up to sign and scale, $A_n(a)$, with $t_L \geq t_R > 0$ and $a = t_R/t_L$. Its reciprocal case is the normal boundary, its one-way limit is the Jordan boundary, and its nonreciprocal interior is the nonnormal regime. Writing $h = \frac{1}{2} \log(t_L/t_R)$, the closed-form tridiagonal Toeplitz eigenvectors give the right-eigenvector envelope $e^{-h(j-1)}$ [17]; analogous exponential edge localization of the nontrivial modes is proved for a finite gauge-capacitance chain in [2]. The same exponent appears in

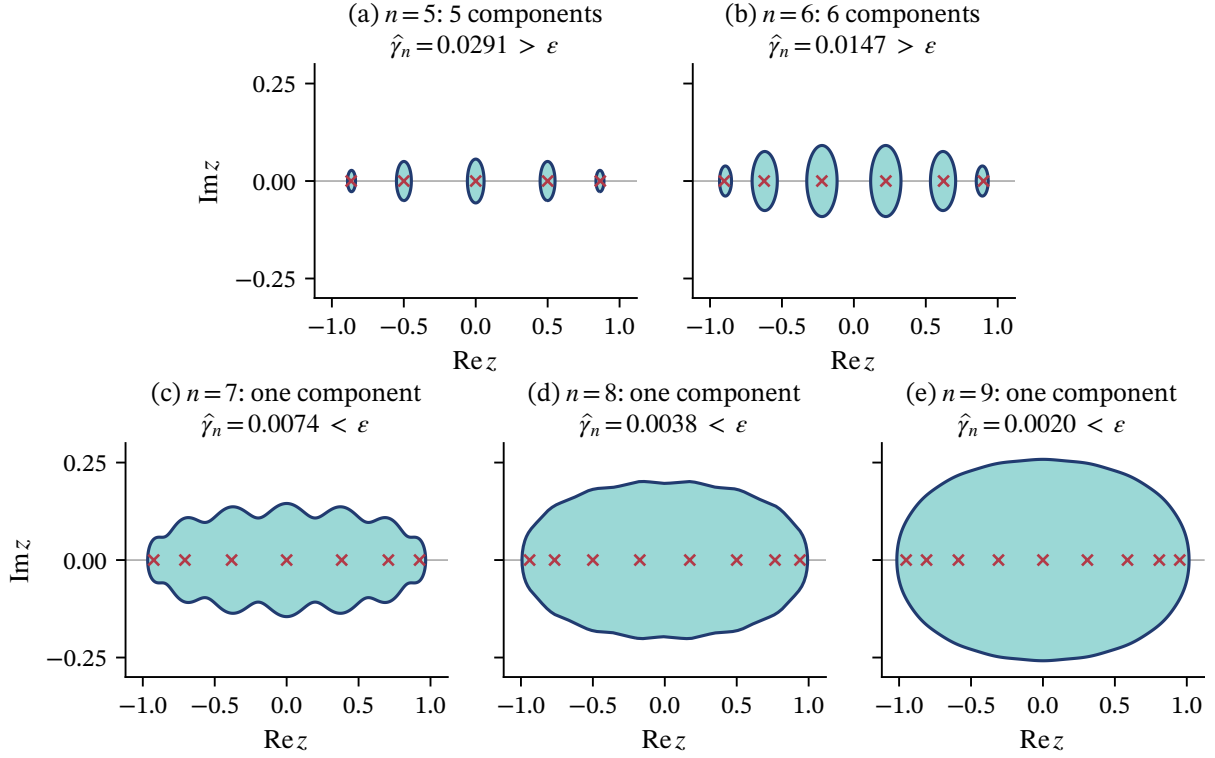


Figure 1: Numerically sampled open pseudospectra of $A_n(1/4)$ at $\varepsilon = 10^{-2}$. Shading marks the sampled pseudospectrum and crosses mark the exact eigenvalues. The samples show component coalescence between $n = 6$ and $n = 7$.

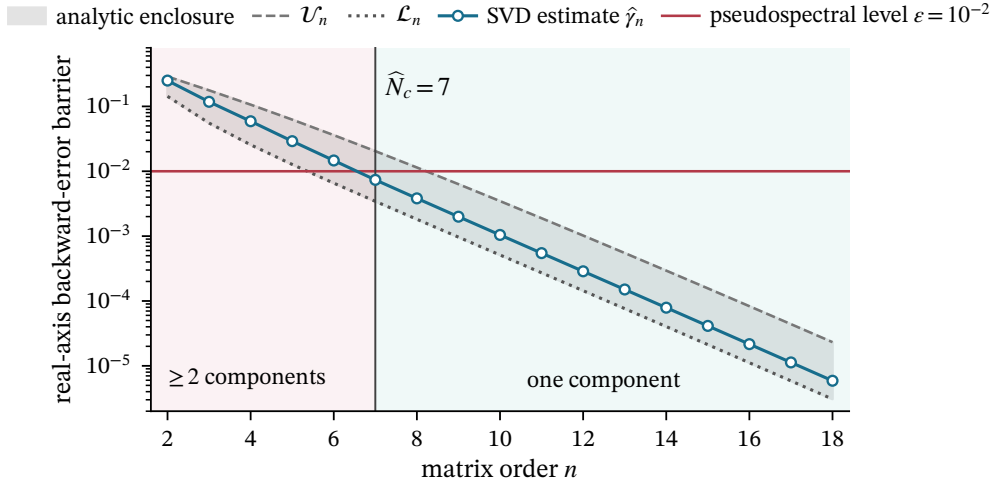


Figure 2: Real-axis backward-error barriers for $a = 1/4$. The gray band, bounded by the dotted lower curve $\mathcal{L}_n(a)$ and dashed upper curve $\mathcal{U}_n(a)$, is the rigorous enclosure of γ_n and establishes the order $(n+1)a^{n/2}$. Blue circles joined by a solid line are floating-point estimates; the red horizontal line is the fixed pseudospectral level $\varepsilon = 10^{-2}$, and the black vertical line marks the numerical crossing $\hat{N}_c = 7$. The theorem gives the bracket $6 \leq N_c \leq 9$.

$a^{n/2} = e^{-hn}$. Thus eigenvector localization and pseudospectral connectedness share the same exponential scale in the nonreciprocal chain.

A Pentadiagonal continuant certificate for vertical monotonicity

Proposition A.1 (Vertical determinant certificate). *Fix $0 < a < 1$, $x \in \mathbb{R}$, $Y \geq 0$, and $t \in \mathbb{R}$, and extend the matrix notation by $A_1 = [0]$. For $k \geq 1$, set*

$$\begin{aligned}\mathcal{P}_k(Y, t) &= ((x + i\sqrt{Y})I_k - A_k)^*((x + i\sqrt{Y})I_k - A_k) - t^2 I_k, \\ F_k(Y, t) &= \det \mathcal{P}_k(Y, t),\end{aligned}$$

and set $F_0 = 1$. For every $n \geq 1$,

$$\mathcal{P}_n(Y, t) > 0 \implies \partial_Y F_n(Y, t) > 0.$$

Proof. At a fixed value of Y , write

$$\delta_{\text{pent}} = x^2 + Y + 1 + a^2 - t^2, \quad b = -(1 + a)x + i(1 - a)\sqrt{Y}.$$

For $n \geq 2$, the matrix $\mathcal{P}_n(Y, t)$ has diagonal

$$\delta_{\text{pent}} - 1, \delta_{\text{pent}}, \dots, \delta_{\text{pent}}, \delta_{\text{pent}} - a^2,$$

whereas for $n = 1$ its sole entry is $\delta_{\text{pent}} - 1 - a^2$. In all dimensions its first upper diagonal is b , its first lower diagonal is $\bar{b}$, and its second off-diagonals are a . Let T_k be the determinant of the pure pentadiagonal Toeplitz matrix with diagonal δ_{pent} and these two off-diagonals. Set $T_{-1} = 0$, $T_0 = 1$, and put

$$D_k = T_k - T_{k-1}, \quad R_k = T_k - a^2 T_{k-1}. \quad (\text{A.1})$$

Since a determinant is affine in each diagonal entry, replacing the first diagonal entry δ_{pent} by $\delta_{\text{pent}} - 1$ subtracts the cofactor T_{k-1} , whereas replacing the last one by $\delta_{\text{pent}} - a^2$ subtracts $a^2 T_{k-1}$. Thus, for $k < n$, D_k and R_k are, respectively, the leading and trailing principal minors of $\mathcal{P}_n(Y, t)$.

The following continuant calculation isolates all dependence on Y . With $T_k = 0$ for $k < 0$, a Laplace expansion over the last two rows, grouped according to the largest displaced index in the contributing permutation, gives for every $k \geq 1$

$$\begin{aligned}T_k &= \delta_{\text{pent}} T_{k-1} - (|b|^2 - a^2) T_{k-2} - (2a^2 \delta_{\text{pent}} - a(b^2 + \bar{b}^2)) T_{k-3} \\ &\quad - a^2 (|b|^2 - a^2) T_{k-4} + a^4 \delta_{\text{pent}} T_{k-5} - a^6 T_{k-6} - a^2 \mathbf{1}_{\{k=2\}}.\end{aligned} \quad (\text{A.2})$$

For example, $T_1 = \delta_{\text{pent}}$ and $T_2 = \delta_{\text{pent}}^2 - |b|^2$; these also verify the only inhomogeneous coefficient in (A.2). Multiplying the recurrence by z^k and summing gives

$$\sum_{k \geq 0} T_k z^k = \frac{1 - a^2 z^2}{Q(z, Y)}, \quad (\text{A.3})$$

where

$$\begin{aligned}Q(z, Y) &= 1 - \delta_{\text{pent}} z + (|b|^2 - a^2) z^2 + (2a^2 \delta_{\text{pent}} - a(b^2 + \bar{b}^2)) z^3 \\ &\quad + a^2 (|b|^2 - a^2) z^4 - a^4 \delta_{\text{pent}} z^5 + a^6 z^6.\end{aligned}$$

Substituting (A.3) into (A.1), with $\mathcal{D}(z, Y) = \sum_{k \geq 0} D_k z^k$, gives the first identity in

$$\mathcal{D}(z, Y) = \frac{(1 - z)(1 - a^2 z^2)}{Q(z, Y)}, \quad \sum_{k \geq 0} F_k(Y, t) z^k = (1 - a^2 z) \mathcal{D}(z, Y). \quad (\text{A.4})$$

For the second identity in (A.4), first change the initial diagonal, producing D_k , and then change the final diagonal; its cofactor is D_{k-1} . Thus $F_k = D_k - a^2 D_{k-1}$, which proves the asserted generating function. Direct substitution of δ_{pent} and b gives

$$\partial_Y Q = -z(1 - z)(1 + az)^2(1 - a^2 z).$$

Differentiating (A.4) therefore yields

$$\sum_{k \geq 0} \partial_Y F_k(Y, t) z^k = \varphi_a(z) \mathcal{D}(z, Y)^2, \quad \varphi_a(z) = \frac{z(1+az)(1-a^2z)^2}{1-az}. \quad (\text{A.5})$$

Suppose that $\mathcal{P}_n(Y, t) > 0$. Then

$$D_k > 0, \quad R_k > 0 \quad (0 \leq k < n). \quad (\text{A.6})$$

Under the positivity in (A.6), put $u_k = \sum_{j=0}^k D_j D_{k-j}$, with $u_k = 0$ for $k < 0$. To extract coefficients from (A.5), note that the coefficients of φ_a , beginning with the coefficient of z , are

$$1, \quad 2a(1-a), \quad a^2(a^2-4a+2), \quad 2a^{k-1}(1-a)^2 \quad (k \geq 4).$$

Thus

$$\begin{aligned} \partial_Y F_n &= u_{n-1} + 2a(1-a)u_{n-2} + a^2(a^2-4a+2)u_{n-3} \\ &\quad + 2(1-a)^2 \sum_{k=4}^n a^{k-1} u_{n-k}. \end{aligned} \quad (\text{A.7})$$

The third coefficient is the sole term that may be negative; the following cofactor inequality controls its contribution. Let $j, k \geq 1$, $j+k = n-1$. Deleting index $k+1$ from $\mathcal{P}_n$ leaves blocks of sizes k and j , with the second off-diagonal a as their sole coupling. In block form, the deleted principal submatrix is

$$\begin{bmatrix} L_k & a\mathbf{e}_k \mathbf{e}_1^T \\ a\mathbf{e}_1 \mathbf{e}_k^T & R_j^{\text{mat}} \end{bmatrix}, \quad \det L_k = D_k, \quad \det R_j^{\text{mat}} = R_j.$$

Expanding once in each of the two coupling entries shows that its determinant is

$$D_k R_j - a^2 D_{k-1} R_{j-1} > 0. \quad (\text{A.8})$$

Deleting index $j+1$ instead gives the same inequality with j, k interchanged. Define

$$\vartheta_k = \frac{D_k}{D_{k-1}}, \quad \varpi_k = \frac{R_k}{R_{k-1}}.$$

Since $T_k = \sum_{\ell=0}^k D_\ell$, (A.1) gives

$$R_k = D_k + (1-a^2) \sum_{\ell=0}^{k-1} D_\ell, \quad R_k - R_{k-1} = D_k - a^2 D_{k-1}. \quad (\text{A.9})$$

The second identity in (A.9) shows that $\vartheta_k \leq a^2$ implies $\varpi_k \leq 1$. On dividing (A.8) and its companion by positive factors, we obtain

$$\vartheta_k \varpi_j > a^2, \quad \vartheta_j \varpi_k > a^2. \quad (\text{A.10})$$

If both ϑ_j and ϑ_k were at most a^2 , then both ϖ_j and ϖ_k would be at most 1, contradicting (A.10). Hence, after interchanging j, k if necessary, $\vartheta_j > a^2$. Moreover,

$$\vartheta_j R_{j-1} - a^2 R_j = (\vartheta_j - a^2)(R_{j-1} - a^2 D_{j-1}),$$

and the second factor equals $(1-a^2) \sum_{\ell=0}^{j-1} D_\ell > 0$. Therefore

$$\vartheta_j > a^2 \implies \varpi_j < \vartheta_j / a^2.$$

The first inequality in (A.10) now gives $a^2 < \vartheta_k \varpi_j < \vartheta_j \vartheta_k / a^2$, and therefore

$$\vartheta_j \vartheta_k > a^4. \quad (\text{A.11})$$

Consequently, for $n \geq 3$, applying (A.11) to $j = p + 1$, $k = n - 2 - p$ gives

$$D_{p+1}D_{n-2-p} > a^4 D_p D_{n-3-p}, \quad 0 \leq p \leq n - 3.$$

Summing and then adding the two positive endpoint terms gives

$$u_{n-1} = 2D_{n-1} + \sum_{j=1}^{n-2} D_j D_{n-1-j} > a^4 u_{n-3}. \quad (\text{A.12})$$

If the third coefficient in (A.7) is negative, write it as $-\mathcal{B}_{\text{neg}}$, where $\mathcal{B}_{\text{neg}} = a^2(4a - 2 - a^2)$. Since

$$a^4 - \mathcal{B}_{\text{neg}} = 2a^2(1 - a)^2 > 0,$$

equations (A.7) and (A.12) prove (4.1). If that coefficient is nonnegative, the implication follows directly. For $n = 1$, one has $\partial_Y F_1 = 1$; for $n = 2$, one has $\partial_Y F_2 = 2D_1 + 2a(1 - a) > 0$. Thus (4.1) holds in every dimension, which proves the proposition. $\square$

B Folded-continuant transfer and boundary residuals

This appendix derives the transfer and generating identities underlying (5.10)–(5.11) and tracks the boundary signs and parity conventions in the signed-pencil argument. Pair the coordinates in the order $(1, n), (2, n - 1), \dots$. In this order the symmetric matrix $(xI_n - A_n)J_n - sI_n$ is block tridiagonal. Its interior diagonal and off-diagonal blocks are

$$D_{\text{int}} = \begin{bmatrix} -s & x \\ x & -s \end{bmatrix}, \quad E_{\text{off}} = \begin{bmatrix} 0 & -1 \\ -a & 0 \end{bmatrix},$$

with lower block E_{off}^T . For even order the terminal block is

$$D_{\text{term}}^e = \begin{bmatrix} -1 - s & x \\ x & -a - s \end{bmatrix};$$

for odd order the terminal block is the scalar $x - s$, coupled to the preceding block by the column $(-1, -a)^T$. Since

$$(xI_n - A_n - sJ_n) = ((xI_n - A_n)J_n - sI_n)J_n,$$

the determinant of this block matrix, multiplied by $\det J_n$, is $q_n(x, s)$. Set $Q_m^e = q_{2m}(x, s)$ and $Q_m^o = q_{2m+1}(x, s)$. The following five-minor transfer completes the continuant verification. For $\tau \in \{e, o\}$, let K_m^τ denote the corresponding matrix

$$(xI_n - A_n - sJ_n)J_n = (xI_n - A_n)J_n - sI_n, \quad n = 2m \text{ or } 2m + 1,$$

after simultaneously permuting its rows and columns into the paired order. Thus K_m^e has order $2m$, K_m^o has order $2m + 1$, and

$$K_1^e = \begin{bmatrix} -1 - s & x \\ x & -a - s \end{bmatrix}, \quad K_1^o = \begin{bmatrix} -s & x & -1 \\ x & -s & -a \\ -1 & -a & x - s \end{bmatrix}.$$

For $m \geq 1$, increasing m by one adds one interior block at the exposed end:

$$K_{m+1}^\tau = \begin{bmatrix} D_{\text{int}} & \mathcal{E}_m^\tau \\ (\mathcal{E}_m^\tau)^T & K_m^\tau \end{bmatrix}, \quad \mathcal{E}_m^\tau = [E_{\text{off}} \quad 0], \quad (\text{B.1})$$

where the zero block has the width required by K_m^τ .

For a matrix whose first two coordinates are exposed, write $K(\widehat{I} \mid \widehat{J})$ for the matrix obtained by deleting rows I and columns J . Suppressing the parity superscript, set

$$\begin{aligned} \mathfrak{d}_m &= \det K_m, & \mathfrak{p}_m &= \det K_m(\widehat{1} \mid \widehat{1}), & \mathfrak{q}_m &= \det K_m(\widehat{2} \mid \widehat{2}), \\ \mathfrak{c}_m &= \det K_m(\widehat{1} \mid \widehat{2}), & \mathfrak{r}_m &= \det K_m(\widehat{1}, \widehat{2} \mid \widehat{1}, \widehat{2}). \end{aligned}$$

The off-principal minor obtained by deleting row 2 and column 1 has the same value $\mathfrak{c}_m$, because K_m is symmetric.

The one-step transfer follows from a Schur complement. On the dense set where both K_m and the tail obtained by deleting its first two coordinates are invertible, let

$$F = \begin{bmatrix} u & w \\ w & v \end{bmatrix}$$

be its Schur complement in K_m , and let δ be the determinant of that tail. Then

$$(\mathfrak{d}_m, \mathfrak{p}_m, \mathfrak{q}_m, \mathfrak{c}_m, \mathfrak{r}_m) = \delta(uv - w^2, v, u, w, 1).$$

Since

$$E_{\text{off}} \text{adj}(F) E_{\text{off}}^T = \begin{bmatrix} u & -aw \\ -aw & a^2v \end{bmatrix},$$

taking the Schur complement of K_m in (B.1) gives

$$\begin{aligned} \mathfrak{d}_{m+1} &= (s^2 - x^2)\mathfrak{d}_m + s(a^2\mathfrak{p}_m + \mathfrak{q}_m) - 2ax\mathfrak{c}_m + a^2\mathfrak{r}_m, \\ \mathfrak{p}_{m+1} &= -s\mathfrak{d}_m - a^2\mathfrak{p}_m, \\ \mathfrak{q}_{m+1} &= -s\mathfrak{d}_m - \mathfrak{q}_m, \\ \mathfrak{c}_{m+1} &= x\mathfrak{d}_m + a\mathfrak{c}_m, \\ \mathfrak{r}_{m+1} &= \mathfrak{d}_m. \end{aligned} \tag{B.2}$$

Both sides are polynomial in x and s . For each fixed $0 < a < 1$, the identities hold on a dense nonsingular set, so the polynomial identity theorem extends them to all x, s , including singular matrix cases.

The five quantities in (B.2) satisfy one linear invariant. Define

$$\ell_m = (1 - a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1 - a)x\mathfrak{c}_m - a(1 - a)\mathfrak{r}_m.$$

Direct substitution into (B.2) gives

$$\ell_{m+1} = -a\ell_m.$$

For the two terminal blocks, direct evaluation gives

$$\begin{aligned} &(\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_e \\ &= ((1 + s)(a + s) - x^2, -a - s, -1 - s, x, 1), \\ &(\mathfrak{d}_1, \mathfrak{p}_1, \mathfrak{q}_1, \mathfrak{c}_1, \mathfrak{r}_1)_o \\ &= (x(1 + a^2) + 2as - C_0(x - s), \\ &\quad s^2 - sx - a^2, s^2 - sx - 1, x^2 - sx - a, x - s). \end{aligned}$$

Thus $\ell_1 = 0$ in both parity classes, and consequently

$$(1 - a)\mathfrak{d}_m - as\mathfrak{p}_m + s\mathfrak{q}_m + (1 - a)x\mathfrak{c}_m - a(1 - a)\mathfrak{r}_m = 0 \quad (m \geq 1). \tag{B.3}$$

Because

$$\det J_{2m} = \det J_{2m+1} = (-1)^m,$$

the first coordinate of the following signed state is the required determinant:

$$\boldsymbol{\eta}_m^\tau = (-1)^m [\mathfrak{d}_m^\tau \quad \mathfrak{p}_m^\tau \quad \mathfrak{q}_m^\tau \quad \mathfrak{c}_m^\tau]^T, \quad (\boldsymbol{\eta}_m^e)_1 = Q_m^e, \quad (\boldsymbol{\eta}_m^o)_1 = Q_m^o.$$

Eliminating $\mathfrak{r}_m$ with (B.3) reduces (B.2) to the four-dimensional transfer

$$\boldsymbol{\eta}_{m+1}^\tau = \mathsf{T}_{\text{fold}} \boldsymbol{\eta}_m^\tau, \quad \mathsf{T}_{\text{fold}} = \begin{bmatrix} x^2 - s^2 - a & \frac{a^3 s}{1-a} & -\frac{s}{1-a} & ax \\ s & a^2 & 0 & 0 \\ s & 0 & 1 & 0 \\ -x & 0 & 0 & -a \end{bmatrix}. \quad (\text{B.4})$$

Set

$$\boldsymbol{\eta}_0^e = [1 \quad a^{-1} \quad 1 \quad 0]^T, \quad \boldsymbol{\eta}_0^o = [x - s \quad 1 \quad 1 \quad -1]^T. \quad (\text{B.5})$$

These values incorporate the terminal conditions at $m = 0$. Direct multiplication in (B.4) reproduces the signed terminal minors above. Hence $\boldsymbol{\eta}_m^\tau = \mathsf{T}_{\text{fold}}^m \boldsymbol{\eta}_0^\tau$ for every $m \geq 0$.

A direct determinant calculation gives

$$\begin{aligned} \det(\lambda I_4 - \mathsf{T}_{\text{fold}}) &= \lambda^4 - C_0 \lambda^3 + \mathfrak{b} \lambda^2 - a^2 C_0 \lambda + a^4, \\ \mathfrak{b} &= (1 + a^2)x^2 + 2as^2 + 2a^2 - 2a - 2a^3. \end{aligned} \quad (\text{B.6})$$

Apply Cayley–Hamilton with the characteristic polynomial (B.6) to T_{fold} , multiply by $\mathsf{T}_{\text{fold}}^{m-4} \boldsymbol{\eta}_0^\tau$, and extract the first coordinate. For both $Z_m = Q_m^e$ and $Z_m = Q_m^o$, this yields

$$Z_m - C_0 Z_{m-1} + \mathfrak{b} Z_{m-2} - a^2 C_0 Z_{m-3} + a^4 Z_{m-4} = 0, \quad m \geq 4. \quad (\text{B.7})$$

Applying T_{fold} three times to (B.5) gives the initial residuals for the recurrence (B.7):

$$\begin{aligned} \text{Res}_0 &= Z_0, \quad \text{Res}_1 = Z_1 - C_0 Z_0, \quad \text{Res}_2 = Z_2 - C_0 Z_1 + \mathfrak{b} Z_0, \\ \text{Res}_3 &= Z_3 - C_0 Z_2 + \mathfrak{b} Z_1 - a^2 C_0 Z_0, \end{aligned}$$

namely

	Res ₀	Res ₁	Res ₂	Res ₃
$Z_m = Q_m^e$	1	$a(1 - 2\zeta_s)$	$a^2(1 - 2\zeta_s)$	a^3
$Z_m = Q_m^o$	$x - s$	$-[x(1 + a^2) + 2as]$	$a^2(x - s)$	0

Thus, as a formal-power-series identity,

$$\Pi(z) \sum_{m \geq 0} Z_m z^m = \text{Res}_0 + \text{Res}_1 z + \text{Res}_2 z^2 + \text{Res}_3 z^3.$$

For the even sequence the polynomial on the right factors as $(1 + az)(1 - 2a\zeta_s z + a^2 z^2)$; for the odd sequence it is $(x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2$. These factorizations identify the boundary-residual polynomials. Since $\Pi(0) = 1$, division by Π in $\mathbb{R}[[z]]$ gives the formal-power-series identities

$$\sum_{m \geq 0} Q_m^e z^m = (1 + az)(1 - 2a\zeta_s z + a^2 z^2) \Pi(z)^{-1}, \quad (\text{B.8})$$

and

$$\sum_{m \geq 0} Q_m^o z^m = ((x - s) - [x(1 + a^2) + 2as]z + a^2(x - s)z^2) \Pi(z)^{-1}. \quad (\text{B.9})$$

C Divided differences for the folded Chebyshev formulas

The preceding transfer calculation gives two rational generating functions. Coefficient extraction converts them to the divided-difference formulas used in the half-angle chord parametrization. Fix $0 < a < 1$ and set $U_{-1} = 0$. The Chebyshev generating function gives

$$\sum_{m \geq 0} (u - \zeta_s)(U_m(u) + U_{m-1}(u))\omega_{\text{gf}}^m = \frac{(u - \zeta_s)(1 + \omega_{\text{gf}})}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2},$$

and, with $\mathbf{B}_s^{\text{fold}} = [x(1 + a^2) + 2as]/(2a)$,

$$\sum_{m \geq 0} ((x - s)u - \mathbf{B}_s^{\text{fold}})U_m(u)\omega_{\text{gf}}^m = \frac{(x - s)u - \mathbf{B}_s^{\text{fold}}}{1 - 2u\omega_{\text{gf}} + \omega_{\text{gf}}^2}.$$

Let $u, v \in \mathbb{R}$ satisfy the two symmetric relations in (5.9). Taking the corresponding divided difference of the Chebyshev generating identities and putting $\omega_{\text{gf}} = az$ recovers (B.8) and (B.9). Coefficient comparison gives (5.10)–(5.11), including the derivative value when $u = v$. The underlying polynomial identities remain valid when $\Delta_{\text{fold}} < 0$, without choosing a square-root branch.

D Exhaustion of the endpoint-side chord sheet

This appendix proves the chord-sheet exhaustion lemma used in the main comparison.

Proof of Lemma 5.3. Continue the ordered middle root $s_*(x)$ through the open gap. Write $\Delta_*(x) = \Delta_{\text{fold}}(x, s_*(x))$ for its folded discriminant, let $y(x) \leq z(x)$ be the reconstructed nonnegative half variables on the real sheet issuing from the endpoint germ, and set $\theta_{\text{ch}}(x) = \arccos y(x)$. Let C be the set of gap points for which

$$\Delta_*(x) > 0, \quad \frac{j\pi}{K} < \theta_{\text{ch}}(x) < \frac{(j+1)\pi}{K}, \quad u_K^{\text{max}} < z(x) < c_h.$$

On C , the reversible folded determinant identities give $\mathcal{W}_K(y(x)) = \mathcal{W}_K(z(x))$; uniqueness on the endpoint-side outer lobe then gives

$$z(x) = z_K(\mathcal{W}_K(y(x))).$$

Thus C is precisely the selected endpoint-side chord sheet. At the upper gap endpoint, $z = c_h$ and $\theta_{\text{ch}} = j\pi/K$. There $z = c_h > 1 \geq y$, so $\Delta_* > 0$ persists immediately inside the gap. Moreover, lemma 5.2 gives $\text{sgn } s_* = (-1)^j$; the signed chord equation therefore places θ_{ch} in $(j\pi/K, (j+1)\pi/K)$. Since $|s_*| > 0$, the channel identity gives $z < c_h$, while continuity keeps $z > u_K^{\text{max}}$. Hence C is nonempty. The ordered eigenvalues of the continuous real symmetric pencil $B_{K-1}(x)$ are continuous in x , including at an eigenvalue crossing, so the continuation passes through a multiple least singular value. The folded pair $u = 2z^2 - 1$, $v = 2y^2 - 1$ and its reconstructed half variables are continuous, while the sheet inequalities are strict. Thus C is relatively open in the gap.

It is also relatively closed. At a relative closure point the chord identity and all sheet inequalities persist weakly. The possible closure cases are as follows. (1) If $\Delta_*(x) = 0$, then the reconstructed half variables coincide, which is impossible because

$$z \geq u_K^{\text{max}} > \cos \frac{\pi}{K} \geq \cos \frac{j\pi}{K} \geq y.$$

(2) At an interior point of a positive gap, including one adjacent to zero, both half variables retain their signs: $z > 0$, while $y = 0$ would force $x = 0$, outside the open positive half. (3) Neither the outer

variable can reach c_h nor the inner variable a nodal endpoint: in either case the common level is zero, the endpoint-side solution is $z = c_h$, and (5.16) gives $s = 0$, impossible inside a spectral gap. (4) The outer variable remains in its endpoint-side component, since its first exit point would be $u_K^{\max}$, contradicting (5.19). Passage through $z = 1$ changes only the coordinate representation. Hence every relative closure point remains in C . Since the open gap is an interval, C is the entire gap. This proves the first exhaustion direction: every actual gap point lies on the selected endpoint-side chord sheet.

For the converse direction, the reconstructed angle θ_{ch} extends continuously to the closed spectral gap. The exact folded variables at the two endpoints give

$$\theta_{\text{ch}}(\lambda_{K-1,j+1}) = \frac{(j+1)\pi}{K}, \quad \theta_{\text{ch}}(\lambda_{K-1,j}) = \frac{j\pi}{K}.$$

For every $\theta_0 \in (j\pi/K, (j+1)\pi/K)$, the intermediate value theorem therefore supplies an interior gap point x with $\theta_{\text{ch}}(x) = \theta_0$. The preceding classification and uniqueness of the endpoint-side outer solution then give

$$y(x) = \cos \theta_0, \quad z(x) = z_K(\mathcal{W}_K(\cos \theta_0)),$$

so the corresponding abscissa and signed root are exactly those of the prescribed j -chord. Thus every chord parameter produces an actual point of the ordered middle branch. By lemma 5.2, its positive magnitude is $s_{\min}(xI_{K-1} - A_{K-1})$.

When K is odd, the same argument applies to the positive half of the single central gap, continuing from the positive spectral endpoint to $x = 0$. Because $z > 0$, the identity $x = (2r/c_h) \cos \theta z$ forces $\theta = \pi/2$ at $x = 0$. Thus every parameter on that half occurs, and reflection covers the negative half. $\square$

E Scalar estimate for the lower central-gap comparison

This appendix proves (5.34) separately in the regimes $\rho > 1$, $\rho = 1$, and $\rho < 1$.

Put $b = u_+/u_- = 1/a$, and define

$$Q_b = b^2 - 2b\rho + 1, \quad S_b = b^2 + 2b\omega_L + 1.$$

The Chebyshev identity gives $u_-^2 Q_b = 1$, and

$$\mathfrak{D}_L = S_b/Q_b, \quad \Delta_\xi := \rho - \xi = \Delta_\omega Q_b/S_b. \quad (\text{E.1})$$

Indeed, since $u_+ = bu_-$ and $u_-^2 = Q_b^{-1}$,

$$\mathfrak{D}_L = 1 + 2(\rho + \omega_L)u_-u_+ = 1 + \frac{2b(\rho + \omega_L)}{Q_b} = \frac{S_b}{Q_b},$$

which proves the first identity in (E.1); the second follows from (5.33). The ratio U_L/U_{L-1} is strictly increasing above the largest zero of U_{L-1} , and equals 1 at $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Hence $\rho > \rho_{\text{ref}}$. Since $S_b - Q_b = 2b(\rho + \omega_L) > 0$,

$$\xi > \rho_{\text{ref}} - \Delta_\omega > 2\omega_L - 1 > 2\omega_L^2 - 1 = \cos \frac{\pi}{L}.$$

Therefore $f_L(x) = (1+x)U_{L-1}(x)$ is positive, strictly increasing, and strictly log-concave on $[\xi, \rho]$. In particular, $\mathfrak{d}_\rho = (\log f_L)'(\rho) > 0$, and

$$\mathfrak{d}_\rho = \begin{cases} \frac{L(b-\rho)-1}{\rho^2-1}, & \rho \neq 1, \\ \frac{2L^2+1}{6}, & \rho = 1 \end{cases}.$$

The endpoint value follows directly from $U_{L-1}(1) = L$ and $U'_{L-1}(1) = L(L^2 - 1)/3$. Moreover,

$$\log f_L(\rho) - \log f_L(\xi) > \Delta_\xi \mathfrak{d}_\rho, \quad f_L(\xi)^2 < f_L(\rho)^2 e^{-2\Delta_\xi \mathfrak{d}_\rho} < \frac{f_L(\rho)^2}{1 + 2\Delta_\xi \mathfrak{d}_\rho}.$$

Here the first strict inequality is the integral form of strict log-concavity. Since $f_L(\rho)^2 = (1 + \rho)^2/Q_b$ and $\Delta_\xi = \Delta_\omega Q_b/S_b$, substitution into (5.34) reduces it to

$$\mathcal{N} = (\rho + \omega_L)(S_b + 2\Delta_\omega Q_b \mathfrak{d}_\rho) - (1 + \omega_L)(1 + \rho)^2 > 0. \quad (\text{E.2})$$

All three cases use

$$\Delta_\omega = 1 - \cos \frac{\pi}{2L} < \frac{\pi^2}{8L^2} < \frac{5}{4L^2}, \quad \Delta_\omega L < 1. \quad (\text{E.3})$$

Suppose first that $\rho > 1$. Put $u = b - \rho$ and $\Delta_+ = \rho^2 - 1$. Since $\mathfrak{d}_\rho > 0$, $u > 1/L$. Keeping ρ fixed, direct differentiation gives

$$\frac{\partial_b \mathcal{N}}{2(\rho + \omega_L)} = \rho + \omega_L + u - \Delta_\omega L + \frac{\Delta_\omega(3Lu^2 - 2u)}{\Delta_+}.$$

The right-hand side is strictly increasing for $u \geq 1/L$ and is positive at $u = 1/L$ by (E.3). Hence $\mathcal{N}$ exceeds its value at $b = \rho + 1/L$. With $s_0 = \rho + \omega_L$ and $\omega_+ = 1 + \omega_L$, that value is

$$\Upsilon(s_0) = s_0 \left[\left(s_0 + \frac{1}{L} \right)^2 + \omega_+ \Delta_\omega \right] - \omega_+(s_0 + \Delta_\omega)^2.$$

For $s_0 \geq \omega_+$,

$$\Upsilon''(s_0) = 6s_0 + 4/L - 2\omega_+ > 0, \quad \Upsilon'(\omega_+) = 2\omega_L \omega_+ + 4\omega_+/L + 1/L^2 > 0,$$

and

$$\Upsilon(\omega_+) = \omega_+(-2\Delta_\omega + 2\omega_+/L + 1/L^2) > 0.$$

Thus (E.2) holds when $\rho > 1$. If $\rho = 1$, then $b = (L + 1)/L$, $Q_b = L^{-2}$, and direct substitution, using the endpoint value of $\mathfrak{d}_\rho$, gives

$$\mathcal{N} = \Upsilon(1 + \omega_L) + \frac{2(1 + \omega_L)\Delta_\omega}{L^2} \frac{2L^2 + 1}{6} > 0.$$

The first term is positive by the preceding calculation, and every factor in the correction term is positive.

For $\rho < 1$, set

$$v = 1 - \rho, \quad \Delta_b = b - 1, \quad u = \Delta_b + v, \quad \Delta_- = v(2 - v).$$

Because $b > 1$, we have $0 < v < u$. Moreover, $\rho^2 - 1 < 0$, so the positivity of $\mathfrak{d}_\rho$ in its quotient formula forces $L(b - \rho) - 1 < 0$. Since $u = b - \rho$,

$$0 < v < u < 1/L. \quad (\text{E.4})$$

With ρ fixed, write $\mathcal{N}_\rho^<(\cdot)$ for $\mathcal{N}$ viewed as a function of $\Delta_b = b - 1$. A direct expansion of (E.2) gives

$$\mathcal{N}_\rho^<(\Delta_b) - \mathcal{N}_\rho^<(0) = \frac{\Delta_b(\rho + \omega_L)}{\Delta_-} \mathfrak{q}(u),$$

where

$$\mathfrak{q}(u) = \Delta_- [2(2 - \Delta_\omega) + u - v] + 2\Delta_\omega [u + v - L(u^2 + uv + v^2) - L\Delta_-].$$

The function q is a concave quadratic. Moreover,

$$\mathcal{N}_\rho^<(0) = \frac{v}{2-v} \left[(2 - \Delta_\omega)(2 - v)^2 - 2\Delta_\omega^2 - 4\Delta_\omega L(2 - v - \Delta_\omega) \right].$$

The following elementary bounds hold; the second follows from $\rho > \rho_{\text{ref}}$ and $1 - \rho_{\text{ref}} < \pi^2/[2(2L+1)^2] < 5/(2L+1)^2$:

$$\Delta_\omega < \frac{5}{4L^2}, \quad v < \frac{5}{(2L+1)^2} \leq \frac{1}{5}. \quad (\text{E.5})$$

When $L = 2$, the non-strict bound equals $1/5$, while the preceding strict estimate still gives $v < 1/5$. These bounds in (E.5) show that the bracketed expression is positive, since it equals

$$(2 - v) \left[(2 - \Delta_\omega)(2 - v) - 4\Delta_\omega L \right] + 2\Delta_\omega^2(2L - 1)$$

and, using $L \geq 2$,

$$(2 - \Delta_\omega)(2 - v) > \frac{27}{16} \frac{9}{5} = \frac{243}{80} > \frac{5}{2} \geq 4\Delta_\omega L.$$

At the endpoints of (E.4),

$$q(v) = v \left[(2 - v)(4 - 2\Delta_\omega) + 4\Delta_\omega - 4\Delta_\omega L(1 + v) \right] > 0,$$

$$q(1/L) = v \left[(2 - v)(4 - 2\Delta_\omega + 1/L - v) - 4\Delta_\omega L \right] > 0.$$

At the two endpoints, the positive leading terms exceed

$$\frac{9}{5} \left(4 - \frac{5}{8} \right) = \frac{243}{40}, \quad \frac{9}{5} \left(4 - \frac{5}{8} - \frac{1}{5} \right) = \frac{1143}{200},$$

while the negative terms have magnitudes less than $(5/L)(6/5) \leq 3$ and $5/L \leq 5/2$, respectively. Concavity therefore gives $q(u) > 0$, proving (E.2), and hence (5.34), in all three cases.

F Signed-root certification in the lower central gap

This appendix verifies the folded substitution and second-singular-value separation used in the main proof of $\mathfrak{h}_m^o > \mathfrak{h}_{m+1}^e$.

At $x = x_*$ and $s = \pm\tau_*$, where $\tau_* = \mathfrak{h}_L^e$, the real pair $u = \xi$, $v = -\omega_L$ satisfies the two symmetric relations in (5.9). Indeed, its sum and product are $\xi - \omega_L$ and $-\omega_L\xi$, and the required identities follow from

$$\mathfrak{D}_L = \frac{\zeta + \omega_L}{\zeta - \rho}, \quad \xi = \frac{\rho(\zeta + 1) + \zeta(\omega_L - 1)}{\zeta + \omega_L}.$$

Positivity of $\xi + \omega_L$ selects the distinct-variable branch of the divided difference in (5.11). Write the folded numerator as

$$\mathfrak{n}_s(t) = -[x(\zeta - t) + s(1 + t)]U_{L-1}(t).$$

Put

$$X_\xi = x_*(\zeta - \xi), \quad Y_\xi = \tau_*(1 + \xi), \quad X_\omega = x_*(\zeta + \omega_L), \quad Y_\omega = \tau_*\Delta_\omega,$$

and set $U_\xi = U_{L-1}(\xi)$, $U_{-\omega} = U_{L-1}(-\omega_L)$. The identities

$$\mathfrak{D}_L = 2a(\zeta + \omega_L)u_+^2, \quad \mathfrak{D}_L(\rho + \omega_L)(\zeta - \xi) = (\zeta + \omega_L)(1 + \xi)$$

give $Y_\omega Y_\xi = X_\omega X_\xi$. The identity $u_-^2 Q_b = 1$ gives $Q_b > 0$, and $S_b - Q_b = 2b(\rho + \omega_L) > 0$, so $\mathfrak{D}_L = S_b/Q_b > 1$. Moreover, $L \geq 2$ gives $\omega_L > 1/2$, while $\rho > \rho_{\text{ref}} > \omega_L$; hence

$$\rho + \omega_L > 2\omega_L > 1 > 1 - \omega_L > 0.$$

Consequently,

$$\frac{X_\omega^2}{Y_\omega^2} = \frac{\mathfrak{D}_L(\rho + \omega_L)}{1 - \omega_L} > 1,$$

so, with $\mathbf{r}_* = X_\xi/Y_\xi = Y_\omega/X_\omega \in (0, 1)$, the common factor $a^{L-1}/(\xi + \omega_L)$ in (5.11) is positive, and the two numerator differences are

$$\begin{aligned} \mathfrak{n}_{\tau_*}(\xi) - \mathfrak{n}_{\tau_*}(-\omega_L) &= (1 + \mathbf{r}_*)(X_\omega U_{-\omega} - Y_\xi U_\xi), \\ \mathfrak{n}_{-\tau_*}(\xi) - \mathfrak{n}_{-\tau_*}(-\omega_L) &= (1 - \mathbf{r}_*)(X_\omega U_{-\omega} + Y_\xi U_\xi). \end{aligned}$$

Their product is positive by (5.34), because

$$U_{-\omega}^2 = \frac{1}{1 - \omega_L^2}, \quad X_\omega^2 U_{-\omega}^2 = \frac{\tau_*^2 \mathfrak{D}_L(\rho + \omega_L)}{1 + \omega_L} > Y_\xi^2 U_\xi^2.$$

Thus

$$q_{2L-1}(x_*, \tau_*) q_{2L-1}(x_*, -\tau_*) > 0.$$

At $x = 0$, the second-smallest singular value of A_{2L-1} is

$$s_{L,2}^{(0)} = \sqrt{1 + a^2 - 2a \cos \frac{\pi}{L}}.$$

Indeed, after separating odd and even coordinates, the two rectangular off-diagonal blocks have Gram matrices equal to the $(L-1)$ -dimensional tridiagonal Toeplitz matrix with diagonal $1 + a^2$ and off-diagonal a . Their singular values therefore have squares

$$1 + a^2 + 2a \cos \frac{k\pi}{L}, \quad 1 \leq k \leq L-1,$$

and each occurs twice, while the remaining singular value is zero. The required separation is

$$x_* + \tau_* < 2\sqrt{a} \sin \frac{\pi}{2L} \leq s_{L,2}^{(0)}.$$

Indeed, recall $\rho_{\text{ref}} = \cos \frac{\pi}{2L+1}$. Since $\rho > \rho_{\text{ref}}$ and U_{L-1} is strictly increasing above its largest zero,

$$u_- = U_{L-1}(\rho) > U_{L-1}(\rho_{\text{ref}}) =: u_{-, \text{ref}}, \quad u_{-, \text{ref}}^{-2} = 2(1 - \rho_{\text{ref}}).$$

The identities $\tau_* = a/u_-$ and $\tau_*^2 = 2a(\zeta - \rho)$ imply $\zeta - \rho < a(1 - \rho_{\text{ref}}) < 1 - \rho_{\text{ref}}$. Also

$$\frac{x_*^2}{2a} = \Delta_\omega \frac{\rho + \omega_L}{\zeta + \omega_L} < \Delta_\omega.$$

Put

$$\theta_L = \frac{\pi}{4L}, \quad \Delta_L = \frac{\theta_L}{2L+1}.$$

Then $\Delta_\omega = 2 \sin^2 \theta_L$ and $1 - \rho_{\text{ref}} = 2 \sin^2(\theta_L - \Delta_L)$. Since all quantities are positive, the preceding quotient bound for x_*^2 gives

$$x_* < 2\sqrt{a} \sin \theta_L.$$

Likewise,

$$\tau_*^2 = 2a(\zeta - \rho) < 2a^2(1 - \rho_{\text{ref}}) = 4a^2 \sin^2(\theta_L - \Delta_L),$$

and hence

$$\tau_* < 2a \sin(\theta_L - \Delta_L) < 2\sqrt{a} \sin(\theta_L - \Delta_L),$$

where the last inequality uses $0 < a < 1$, hence $a < \sqrt{a}$. It suffices to prove

$$\sin \theta_L + \sin(\theta_L - \Delta_L) < \sin(2\theta_L). \quad (\text{F.1})$$

The estimates

$$2 \sin \theta_L - \sin(2\theta_L) = 2 \sin \theta_L (1 - \cos \theta_L) < \theta_L^3$$

and

$$\sin \theta_L - \sin(\theta_L - \Delta_L) = \int_{\theta_L - \Delta_L}^{\theta_L} \cos u \, du > \Delta_L \cos \theta_L$$

reduce (F.1) to $\Delta_L \cos \theta_L > \theta_L^3$. Substituting $\Delta_L = \theta_L/(2L+1)$ and $\theta_L = \pi/(4L)$ gives the equivalent condition $2 \cos \theta_L > \theta_L(\pi + 2\theta_L)$. For $0 < \theta_L \leq \pi/8$, using $\pi^2 < 10$,

$$2 \cos \theta_L > 2 - \theta_L^2 \geq \frac{59}{32} > \frac{25}{16} \geq \theta_L(\pi + 2\theta_L).$$

Consequently $x_* + \tau_* < 2\sqrt{a} \sin \frac{\pi}{2L}$. Finally,

$$(s_{L,2}^{(0)})^2 - 4a \sin^2 \frac{\pi}{2L} = (1-a)^2 \geq 0,$$

which proves (5.37).

Together with the signed product inequality, this proves the signed-root certification used in Section 5.7.

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References

- [1] R. Alam, S. Bora, R. Byers, and M. L. Overton. Characterization and construction of the nearest defective matrix via coalescence of pseudospectral components. *Linear Algebra Appl.*, 435:494–513, 2011. doi:10.1016/j.laa.2010.09.022.
- [2] H. Ammari, S. Barandun, J. Cao, B. Davies, and E. O. Hiltunen. Mathematical foundations of the non-Hermitian skin effect. *Arch. Ration. Mech. Anal.*, 248:33, 2024. doi:10.1007/s00205-024-01976-y.
- [3] H. Ammari, S. Barandun, Y. De Bruijn, P. Liu, and C. Thalhhammer. Spectra and pseudo-spectra of tridiagonal k -Toeplitz matrices and the topological origin of the non-Hermitian skin effect. *J. Phys. A: Math. Theor.*, 58:205201, 2025. doi:10.1088/1751-8121/add5ab.
- [4] H. Ammari, S. Barandun, Y. De Bruijn, P. Liu, and C. Thalhhammer. Corrigendum: Spectra and pseudo-spectra of tridiagonal k -Toeplitz matrices and the topological origin of the non-Hermitian skin effect (2025 J. Phys. A: Math. Theor. 58 205201). *J. Phys. A: Math. Theor.*, 59:079501, 2026. doi:10.1088/1751-8121/ae442e.

- [5] M. Bogoya, J. Gasca, and S. M. Grudsky. Eigenvalues for a class of non-Hermitian tetradiagonal Toeplitz matrices. *J. Spectr. Theory*, 15(1):441–477, 2025. doi:10.4171/JST/538.
- [6] A. Böttcher. Pseudospectra and singular values of large convolution operators. *J. Integral Equations Appl.*, 6(3):267–301, 1994. doi:10.1216/jiea/1181075815.
- [7] A. Böttcher. Spectral instabilities: variations on a theme loved by Brian Davies. *J. Spectr. Theory*, 14(2):459–478, 2024. doi:10.4171/JST/483.
- [8] A. Böttcher and S. M. Grudsky. *Spectral Properties of Banded Toeplitz Matrices*. SIAM, Philadelphia, 2005. doi:10.1137/1.9780898717853.
- [9] J. V. Burke, A. S. Lewis, and M. L. Overton. Optimization and pseudospectra, with applications to robust stability. *SIAM J. Matrix Anal. Appl.*, 25:80–104, 2003. doi:10.1137/S0895479802402818.
- [10] J. V. Burke, A. S. Lewis, and M. L. Overton. Spectral conditioning and pseudospectral growth. *Numer. Math.*, 107(1):27–37, 2007. doi:10.1007/s00211-007-0080-3.
- [11] P. Buttà, N. Guglielmi, and S. Noschese. Computing the structured pseudospectrum of a Toeplitz matrix and its extreme points. *SIAM J. Matrix Anal. Appl.*, 33:1300–1319, 2012. doi:10.1137/120864349.
- [12] S. N. Chandler-Wilde, R. Chonchaiya, and M. Lindner. On spectral inclusion sets and computing the spectra and pseudospectra of bounded linear operators. *J. Spectr. Theory*, 14:719–804, 2024. doi:10.4171/JST/514.
- [13] E. B. Davies. Spectral properties of random non-self-adjoint matrices and operators. *Proc. R. Soc. Lond. A*, 457(2005):191–206, 2001. doi:10.1098/rspa.2000.0662.
- [14] N. Hatano and D. R. Nelson. Localization transitions in non-Hermitian quantum mechanics. *Phys. Rev. Lett.*, 77:570–573, 1996. doi:10.1103/PhysRevLett.77.570.
- [15] N. Hatano and D. R. Nelson. Vortex pinning and non-Hermitian quantum mechanics. *Phys. Rev. B*, 56:8651–8673, 1997. doi:10.1103/PhysRevB.56.8651.
- [16] I. Kiorpelidis and K. G. Makris. Scaling of pseudospectra in exponentially sensitive lattices. *Phys. Rev. Res.*, 7(3):L032043, 2025. doi:10.1103/PhysRevResearch.7.L032043.
- [17] S. Noschese, L. Pasquini, and L. Reichel. Tridiagonal Toeplitz matrices: Properties and novel applications. *Numer. Linear Algebra Appl.*, 20:302–326, 2013. doi:10.1002/nla.1811.
- [18] L. Reichel and L. N. Trefethen. Eigenvalues and pseudo-eigenvalues of Toeplitz matrices. *Linear Algebra Appl.*, 162–164:153–185, 1992. doi:10.1016/0024-3795(92)90374-J.
- [19] P. Schmidt and F. Spitzer. The Toeplitz matrices of an arbitrary Laurent polynomial. *Math. Scand.*, 8:15–38, 1960. doi:10.7146/math.scand.a-10588.
- [20] J. Sirker. Pseudospectral phenomena and the origin of the non-Hermitian skin effect, 2026. arXiv:2603.22643v3. doi:10.48550/arXiv.2603.22643.
- [21] L. N. Trefethen, M. Contedini, and M. Embree. Spectra, pseudospectra, and localization for random bidiagonal matrices. *Comm. Pure Appl. Math.*, 54(5):595–623, 2001. doi:10.1002/cpa.4.
- [22] L. N. Trefethen and M. Embree. *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press, Princeton, NJ, 2005. doi:10.1515/9780691213101.